

EDA Credit Assignment

By
Priyanka Agarwala

PROBLEM STATEMENT

"The loan company wants to lend money to people who can pay it back. But some people don't pay back their loans, which causes problems. I work for a company that lends money to city people. My job is to look at the data about loan applications and see if I can find any patterns.

When someone asks for a loan, the company has to decide if they should give it to them or not. There are two risks:

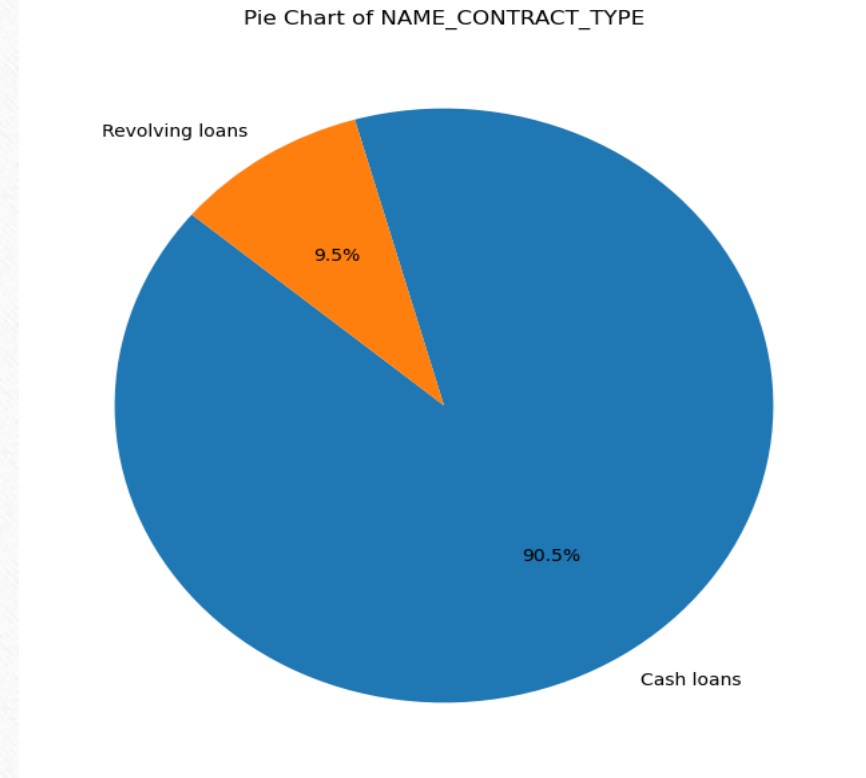
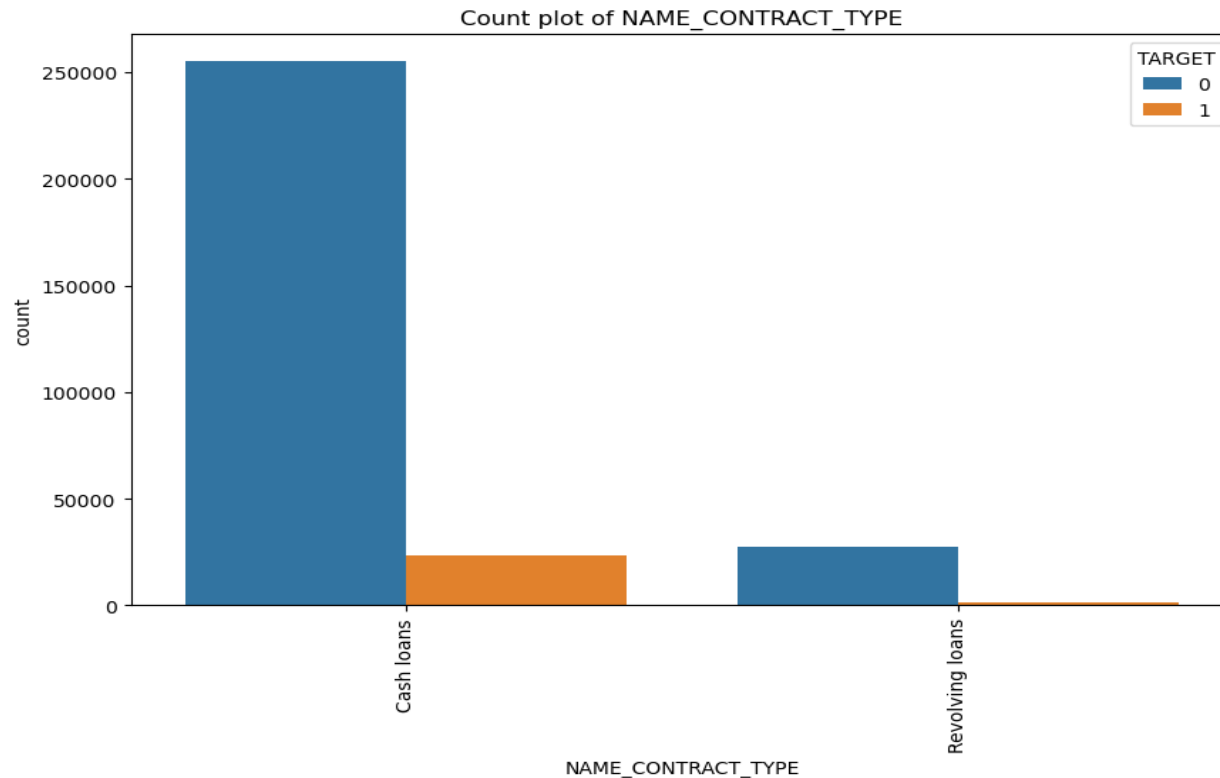
1. If the person can pay back the loan but the company says no, they lose business.
2. If the person can't pay back the loan but the company says yes, they lose money.

This case study is about figuring out the signs that show if someone might struggle to pay back their loan. By analyzing data, the company wants to spot these signs early to make better decisions, like adjusting the loan amount or interest rates. This helps them to lend to people who can pay back the loan while minimizing the risk of defaults. So, basically trying to find out what factors make someone more likely to not pay back a loan on time.

APPROACH AND METHODOLOGY

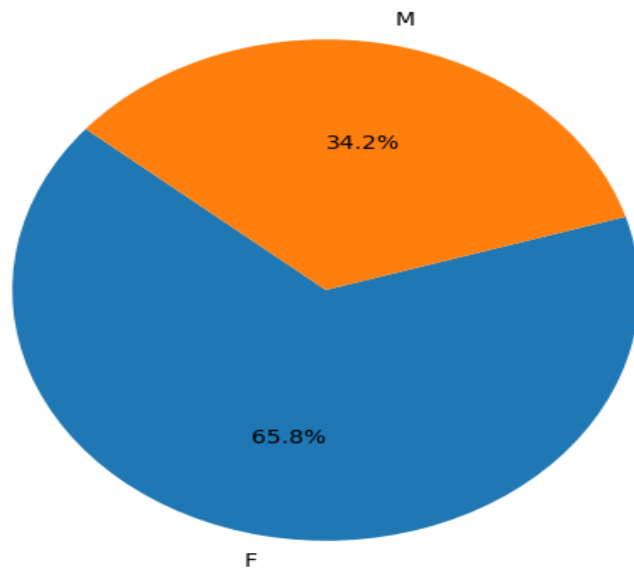
- ☐ Data loading
- ☐ Data understanding
- ☐ Data cleaning (Missing value count, Imputing missing values, Standardizing and binning)
- ☐ Checking Imbalance
- ☐ Outlier handling
- ☐ univariate Analysis (Descriptive Statistics and Visualization)
- ☐ Bi-variate Analysis (Descriptive Statistics and Visualization)

ANALYSIS FOR CURRENT APPLICATION DATA



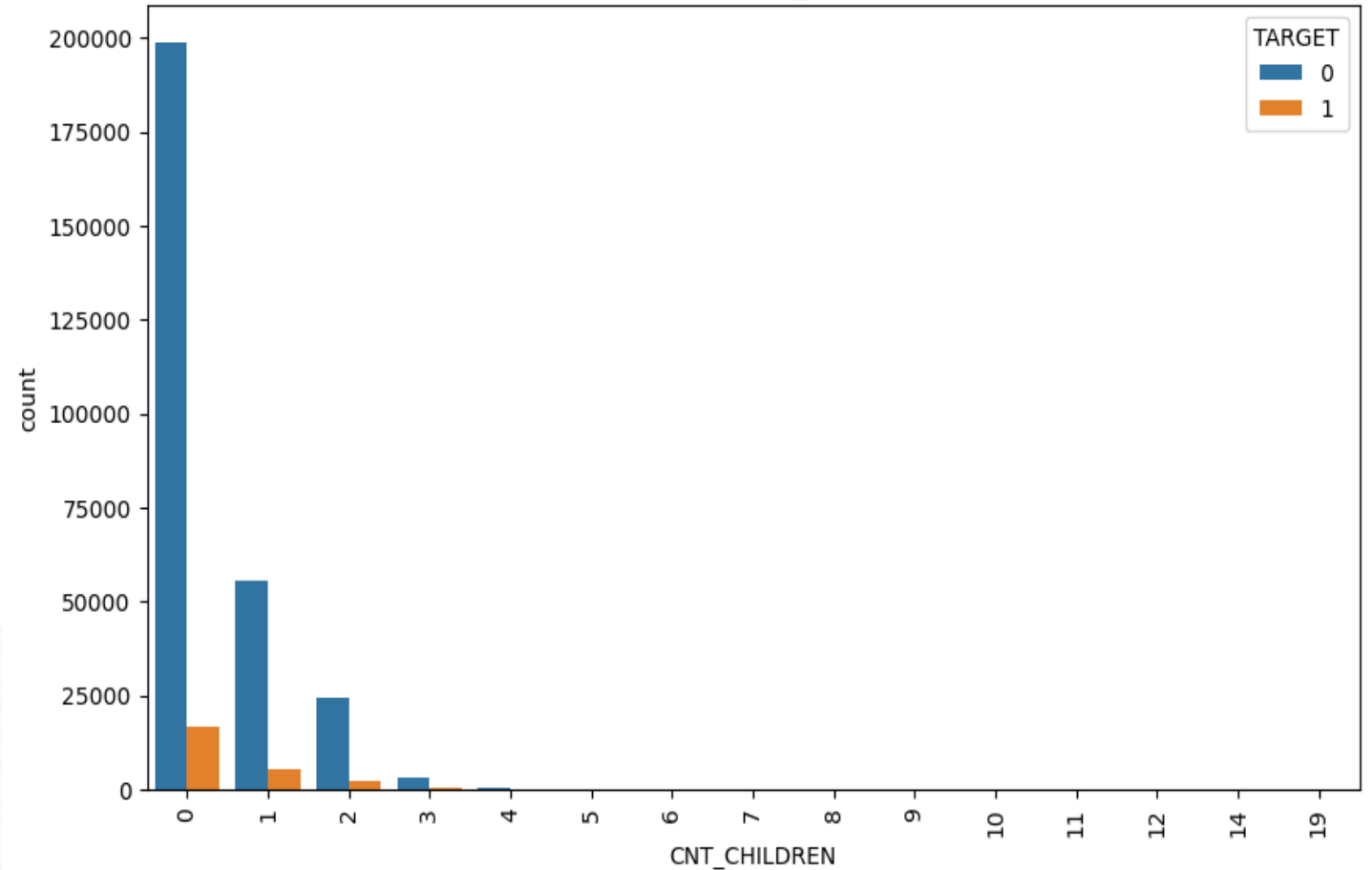
- People tends to take more cash loans as compared to revolving loans. And default percentage of revolving loans are less.

Pie Chart of CODE_GENDER

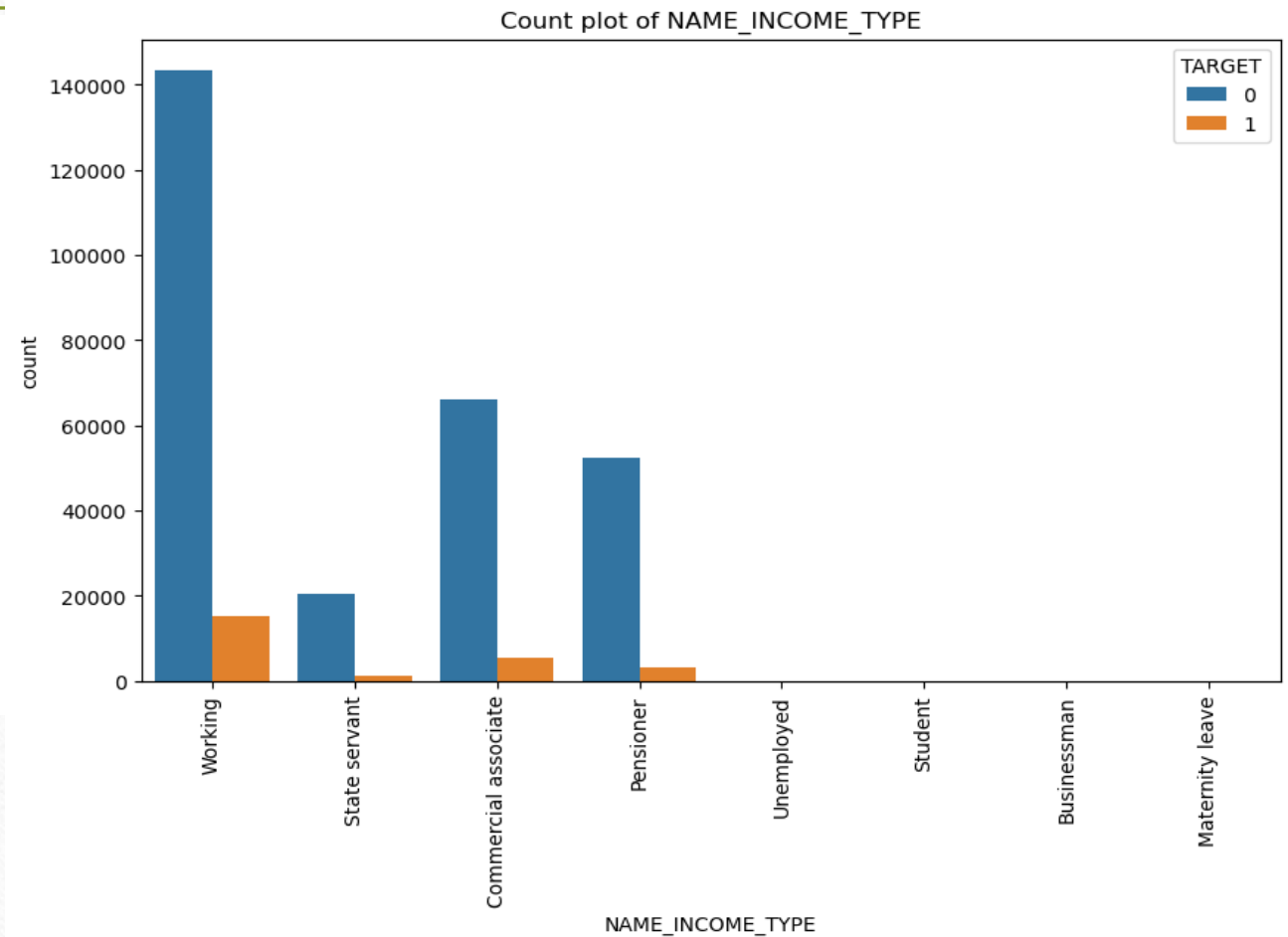
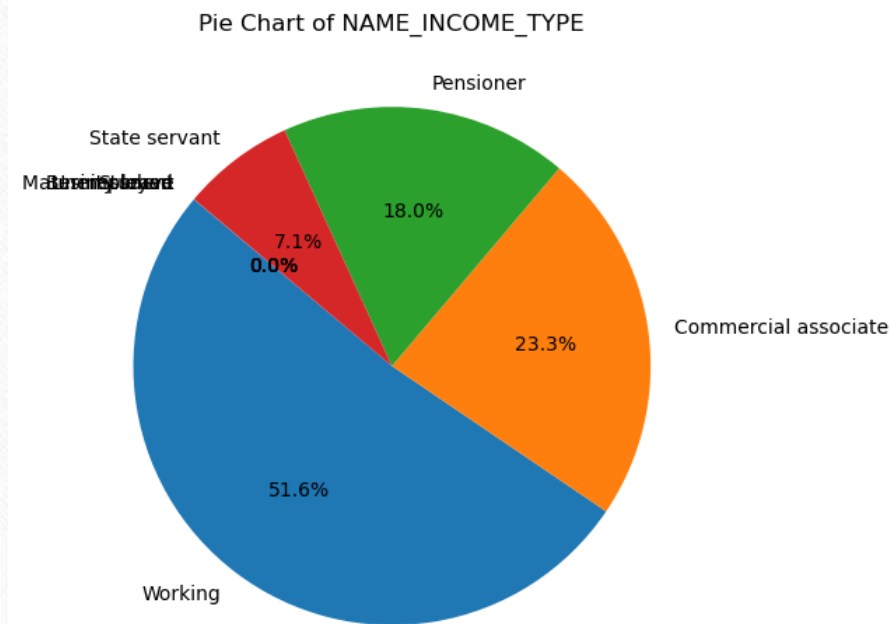


- Females tends to take more loans.

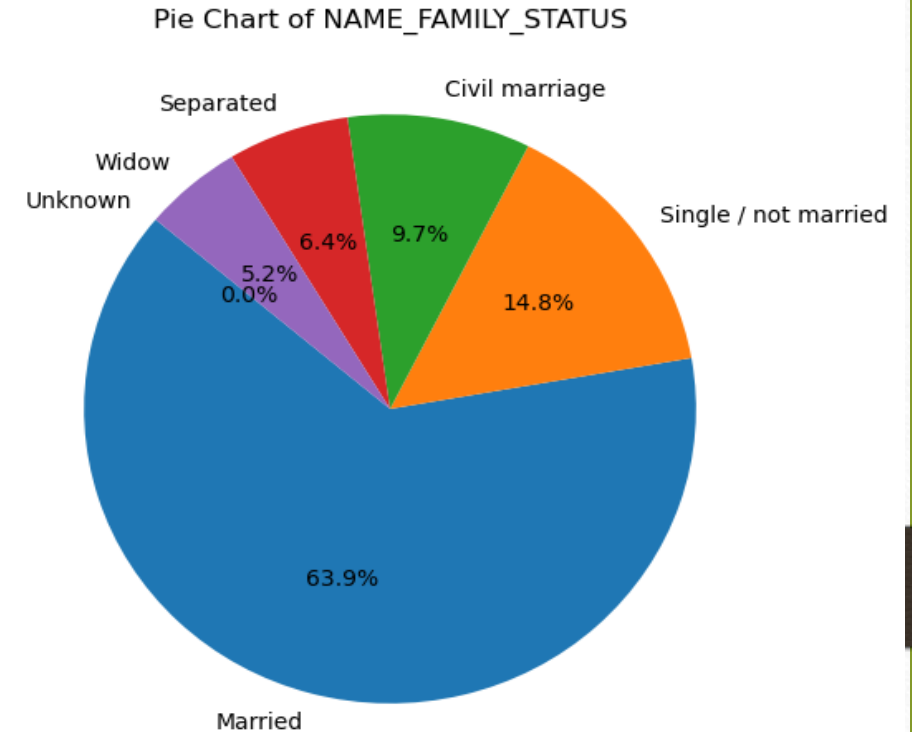
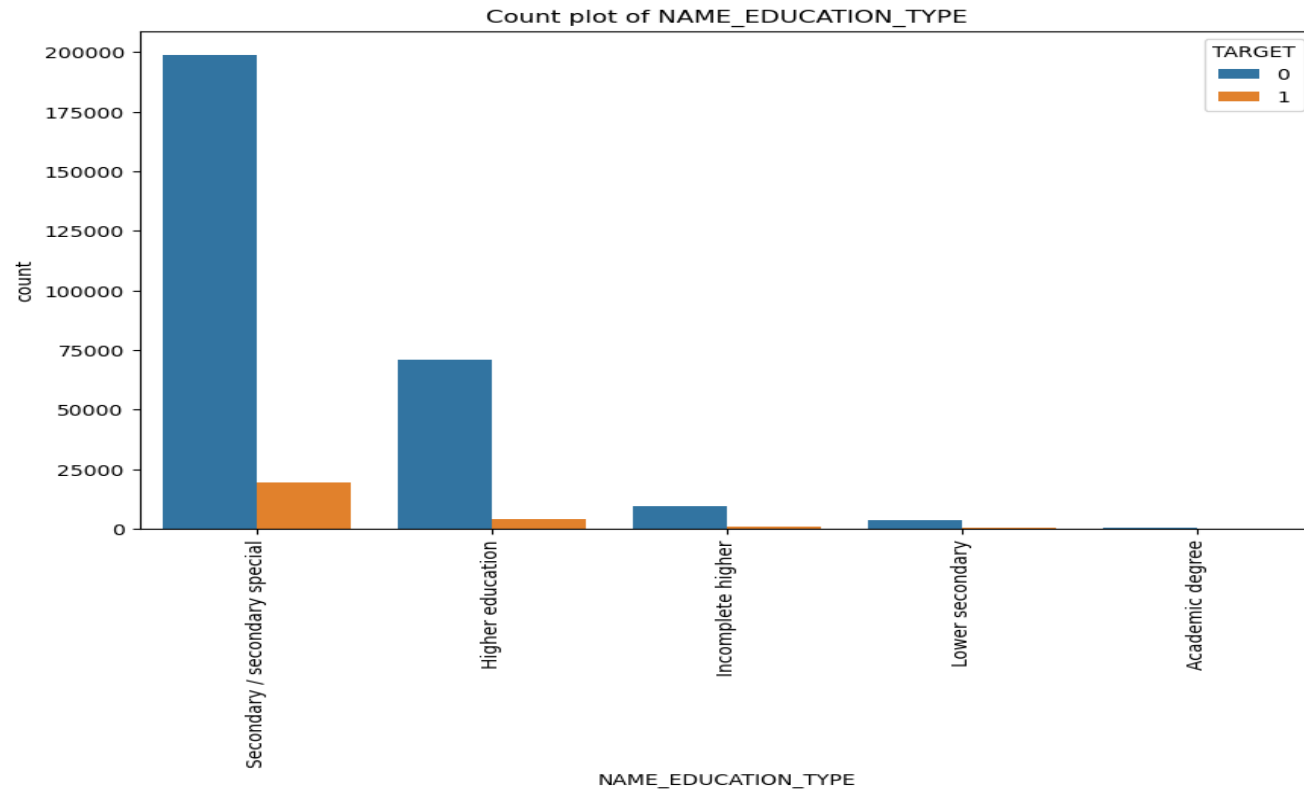
Count plot of CNT_CHILDREN



- Peoples having no child tend to have less default percentage.

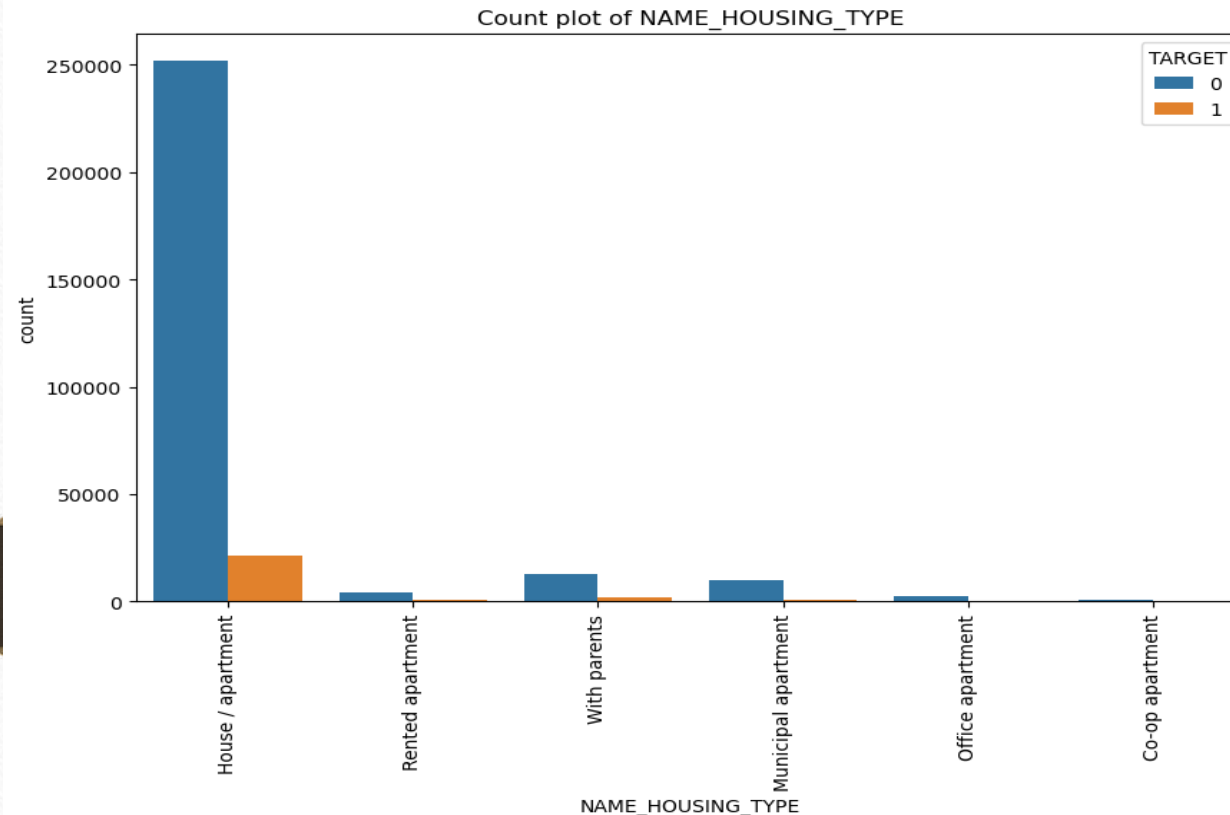


➤ Working class people tends to take more loans.

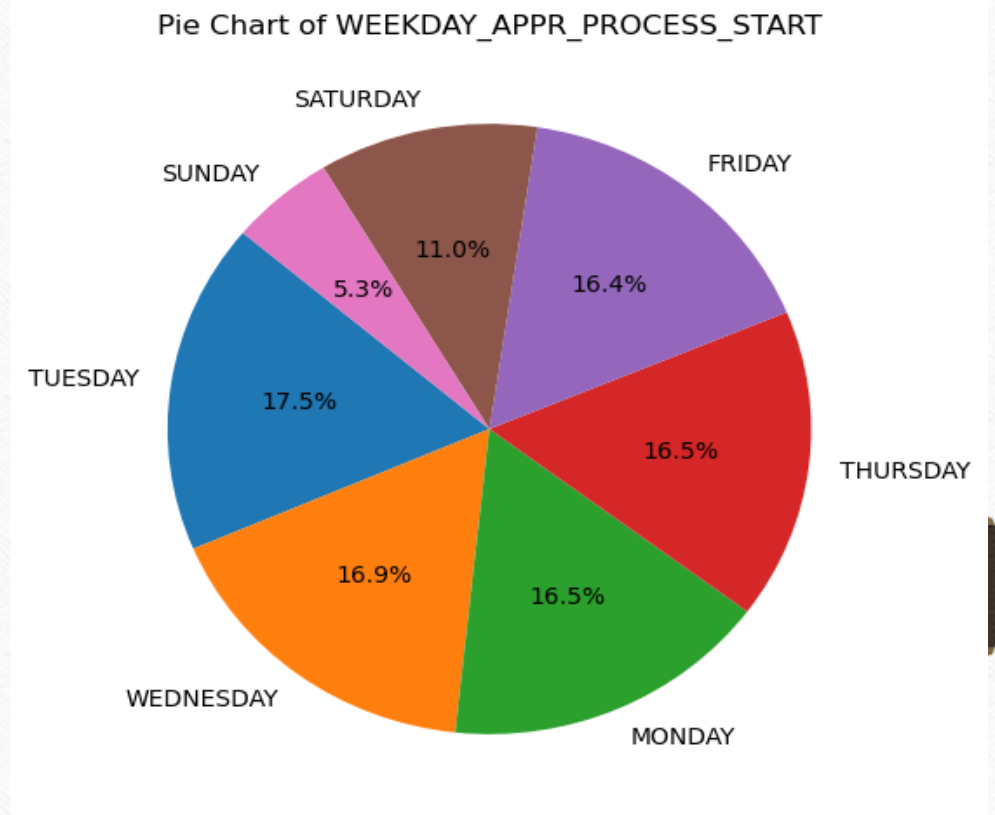


➤ Maximum number of loans are taken by peoples who have education upto Secondary/secondary special .

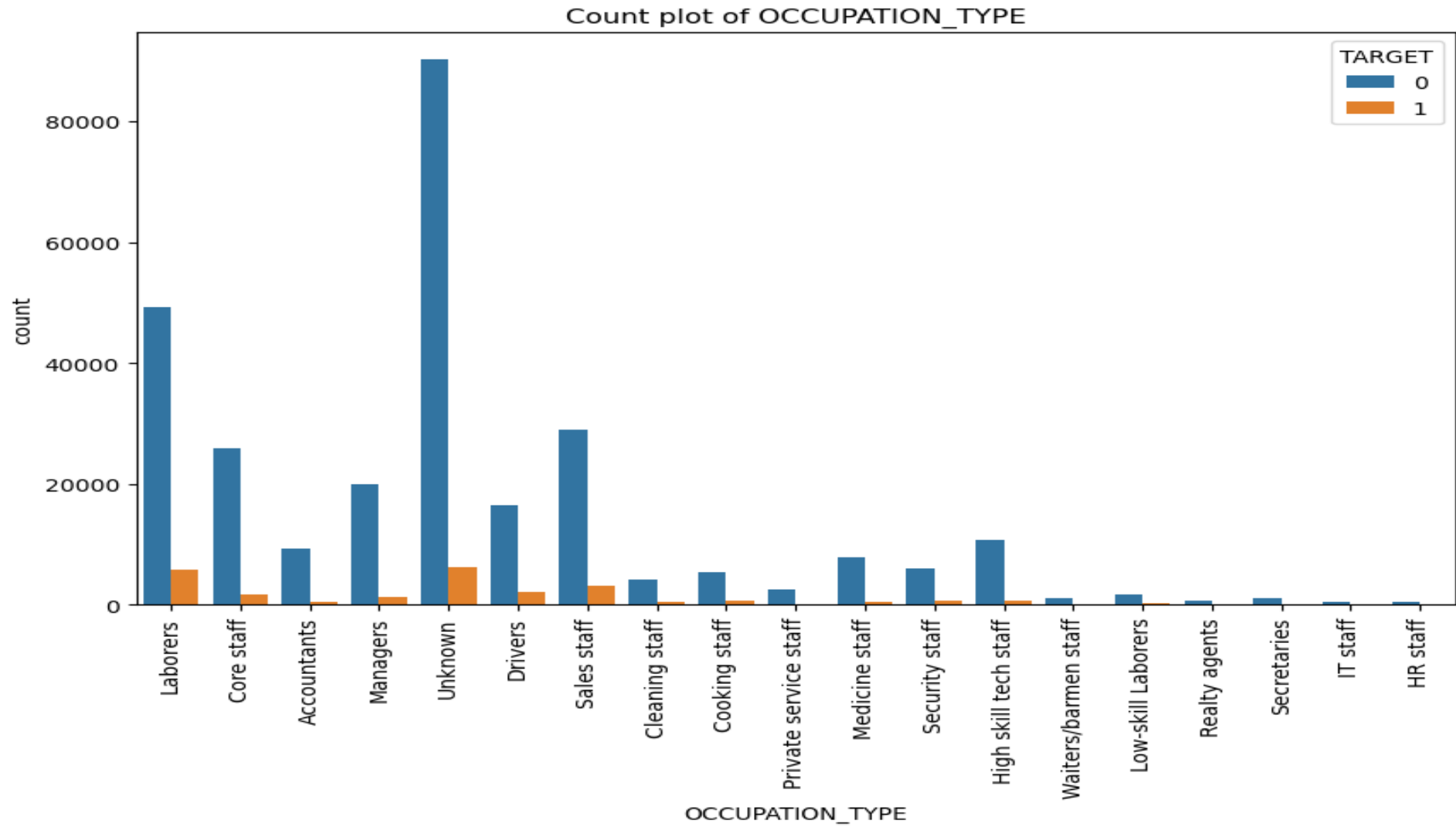
➤ Maximum number of loans are taken by married peoples as compared to other categories.



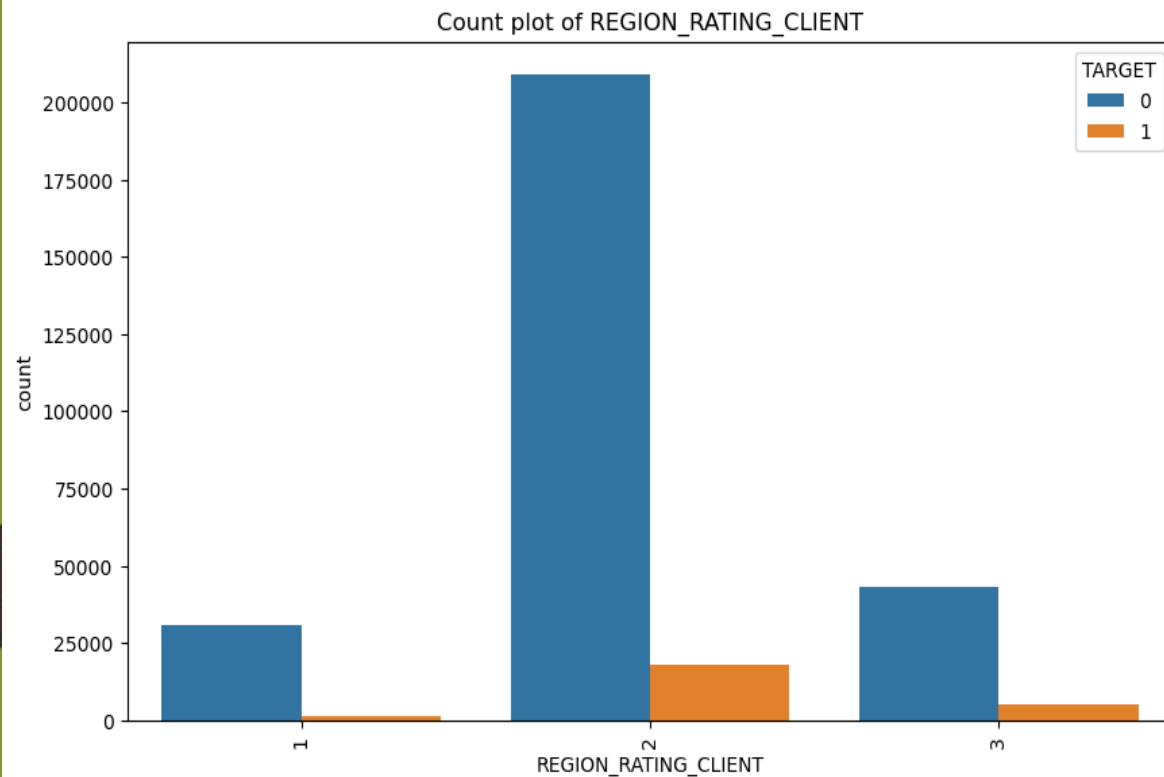
- People with house or apartment tend to take more loans.



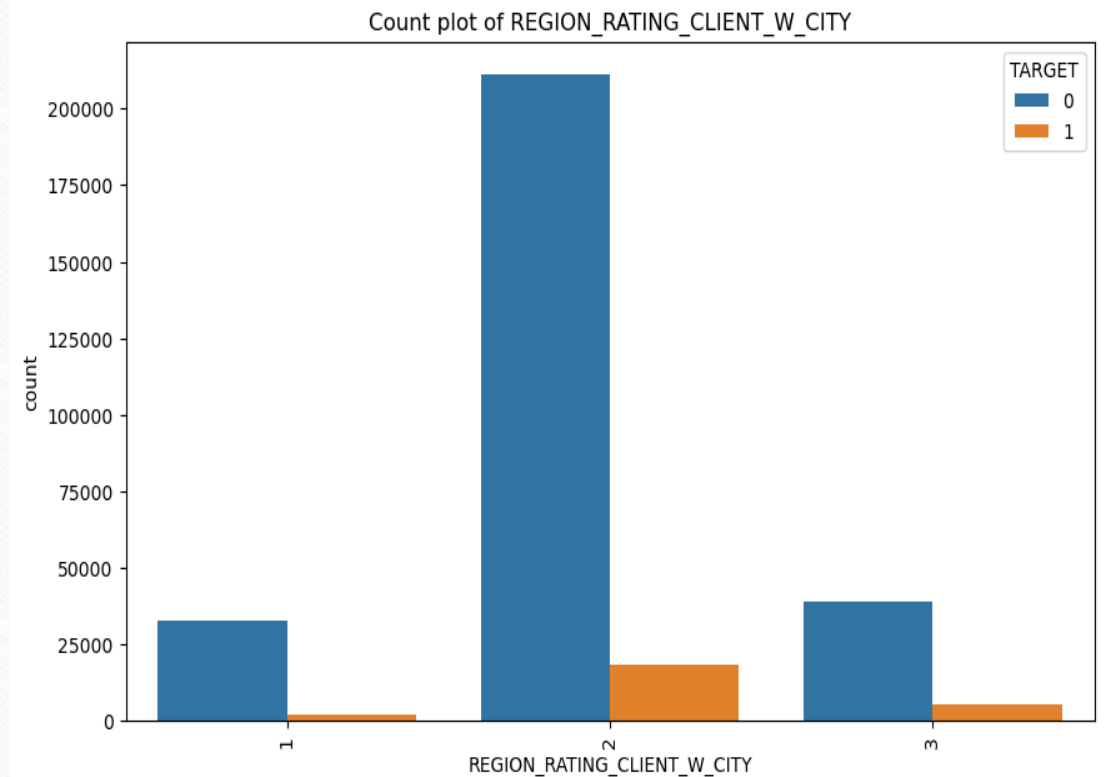
- Saturday and Sunday are less busy for bank in terms of loan applications.
- Tuesday is the most busiest day.
- Sunday is the least busiest day.



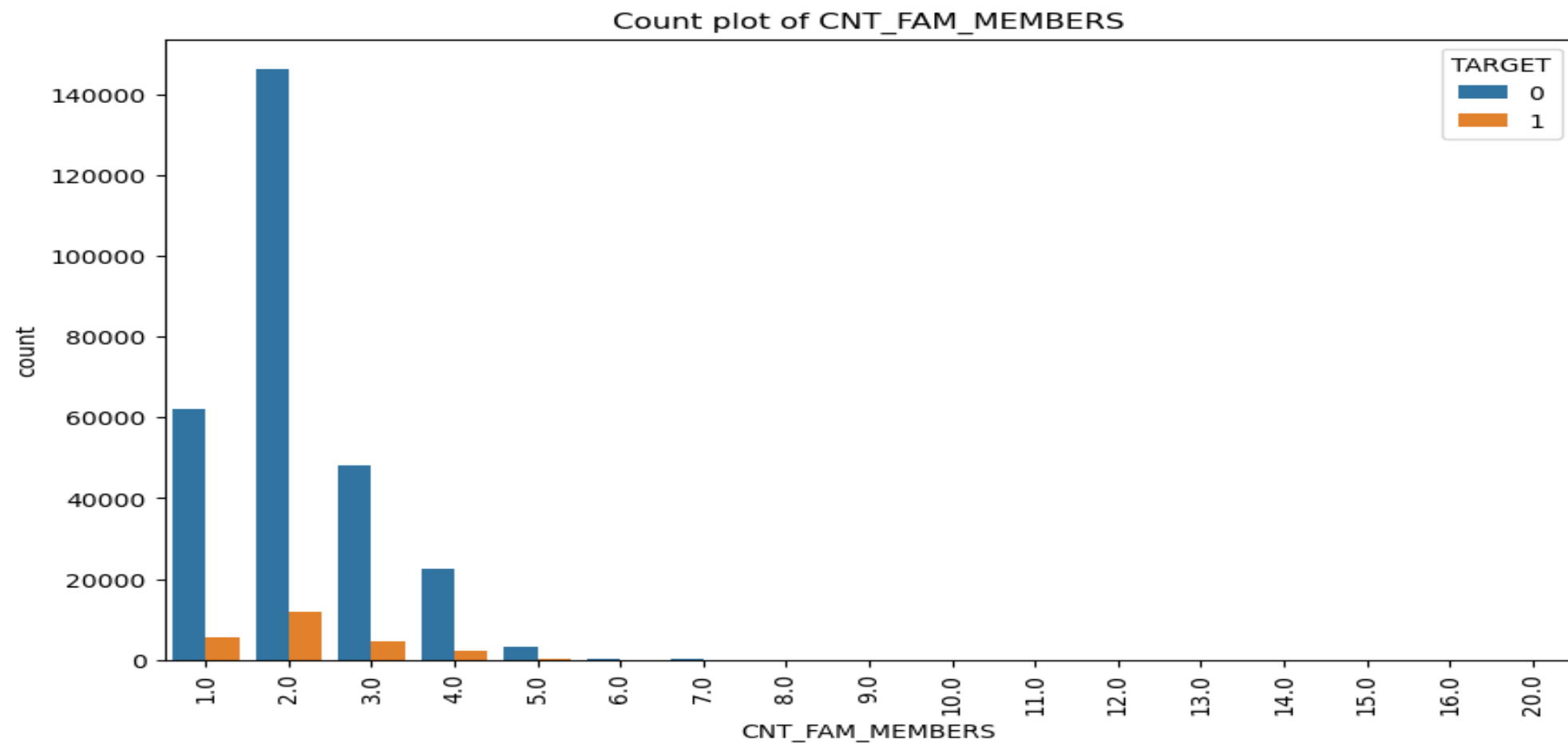
➤ People working as laborers tends to take more loans.



- Maximum number of non-defaulters are the peoples living in region with rating 2 and they are also the maximum loan holders.

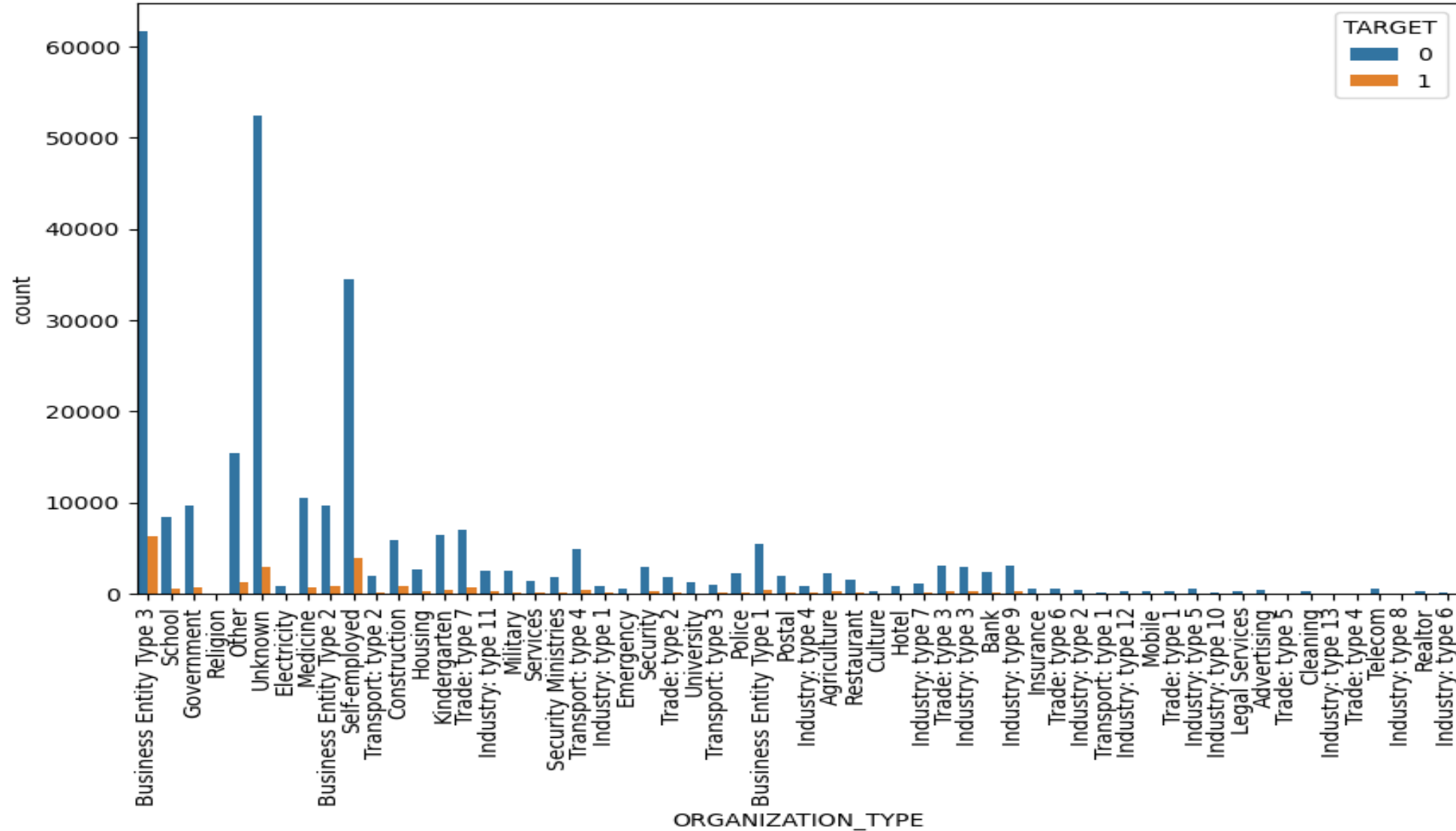


- Maximum number of non-defaulters are the peoples living in region within city with rating 2 and they are also the maximum loan holders.

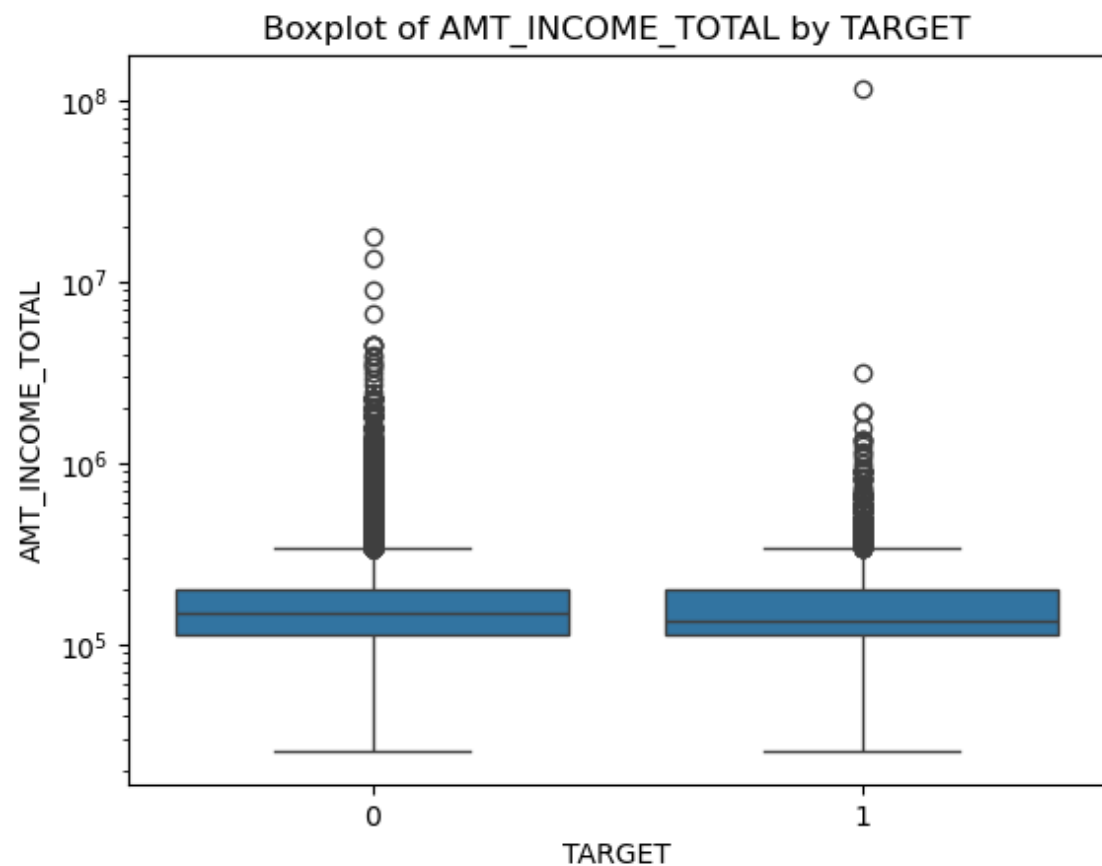


➤ Nuclear families tends to take more loans.

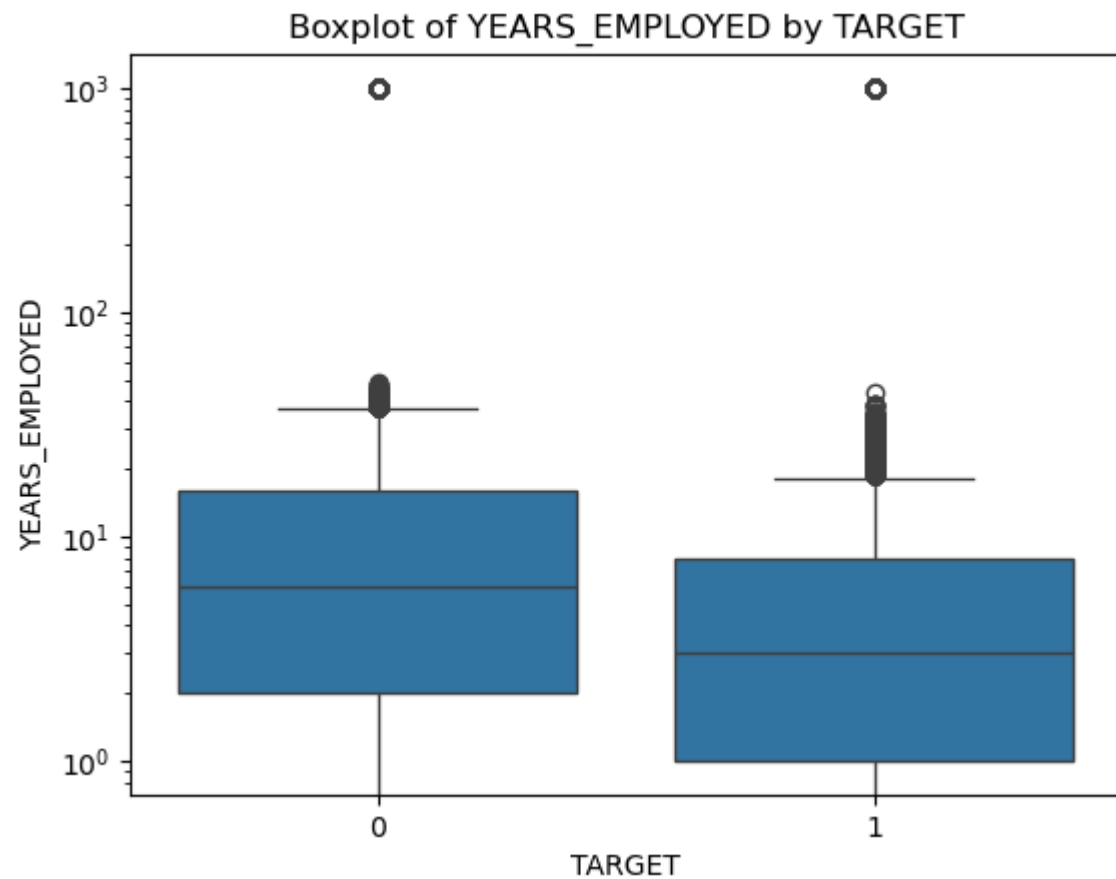
Count plot of ORGANIZATION_TYPE



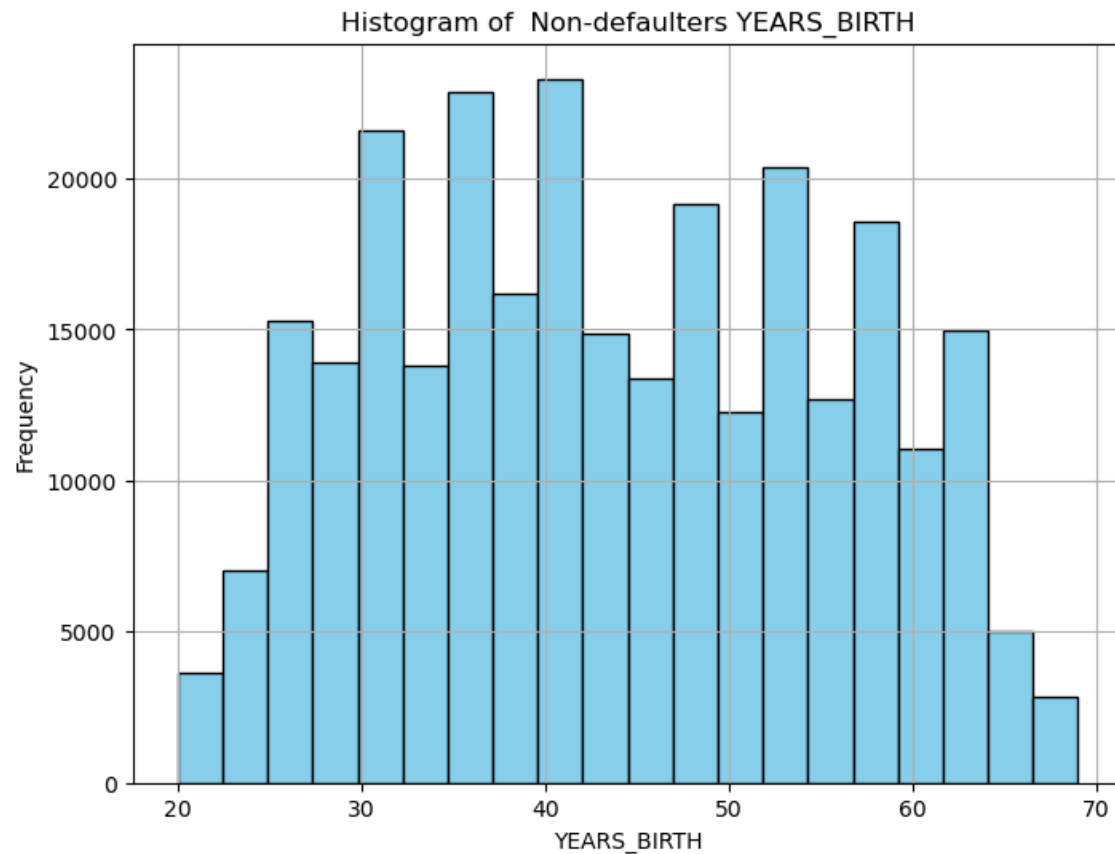
➤ People working in Business entity type 3 tends to take more loans as compared to other categories.



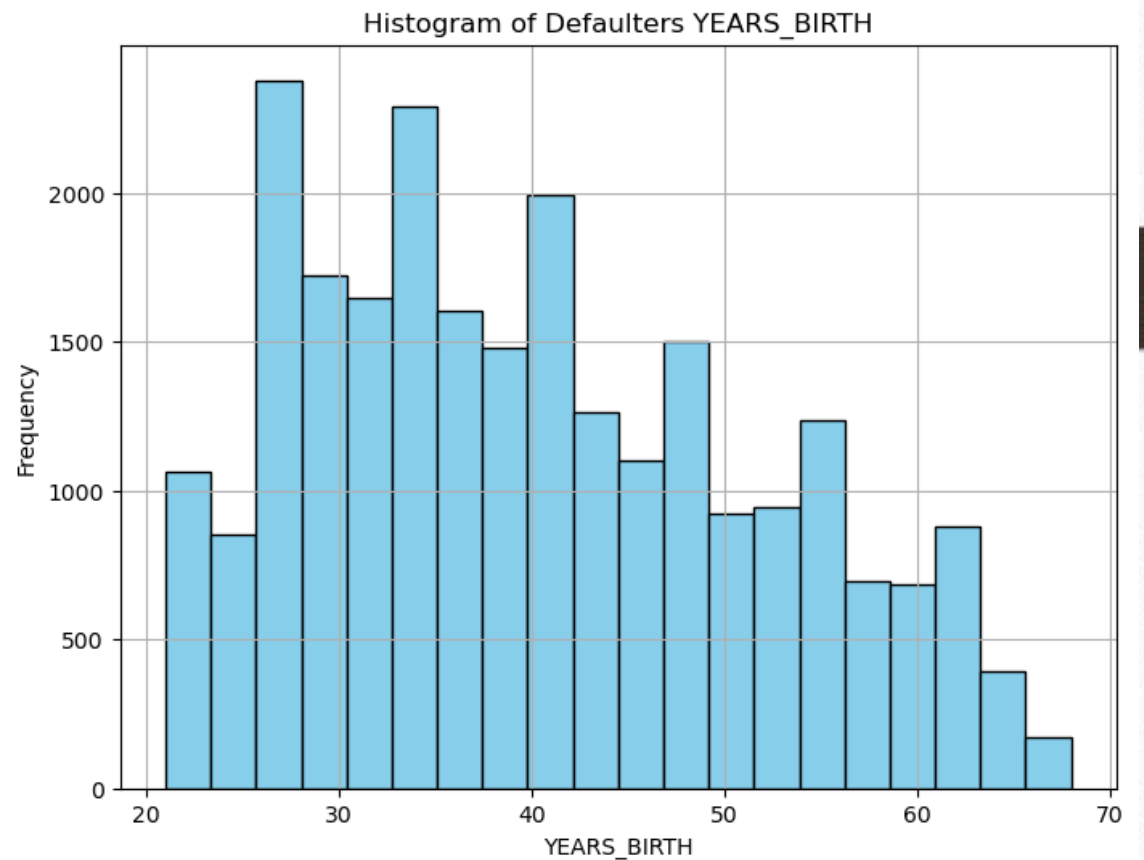
- People with lower total income are more likely to default.

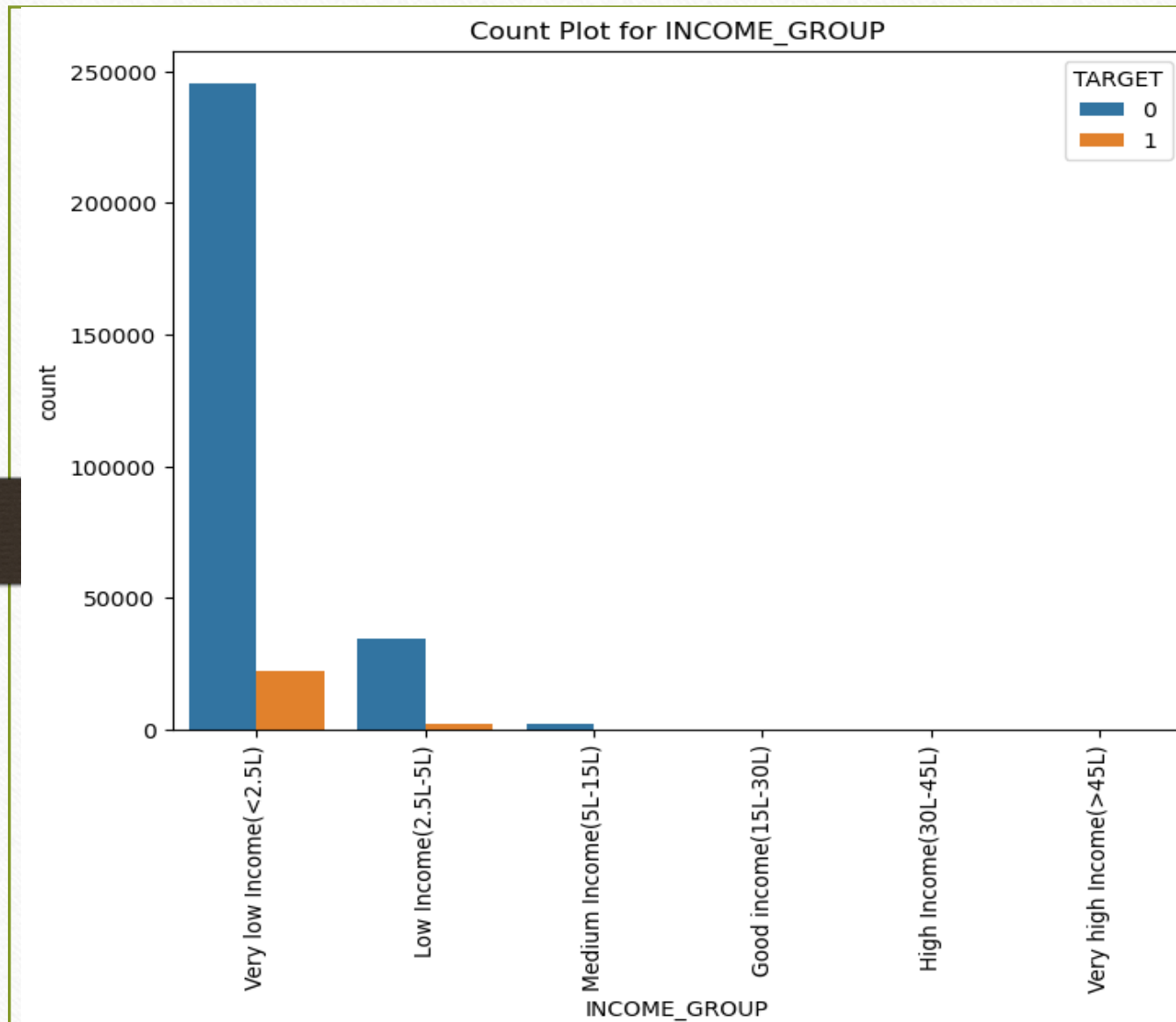


- People with higher number of employment years are less likely to default.

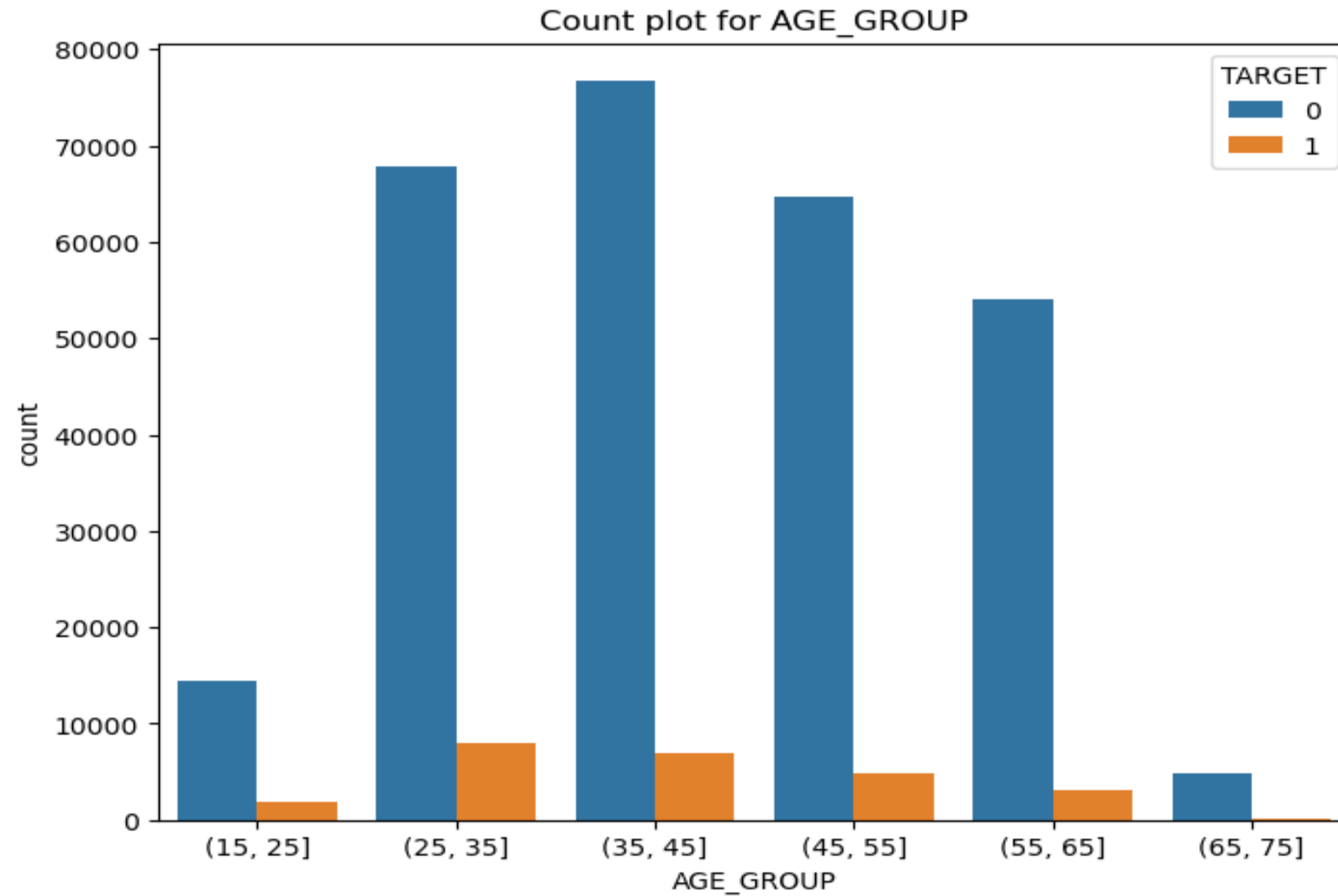


➤ People who retired tends to take more loans and are less defaulters.





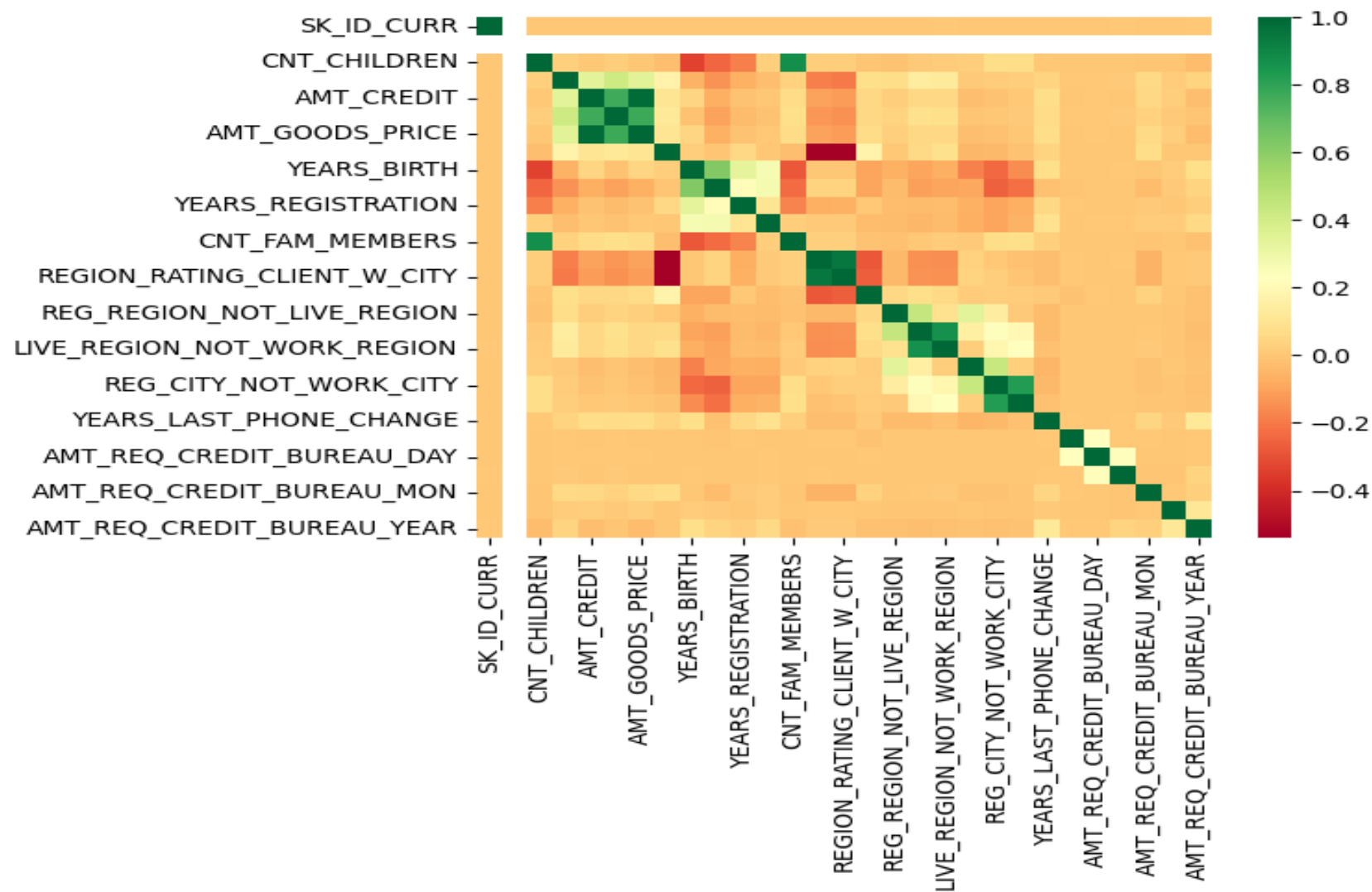
➤ People with lower total income tends to take more loans.



- People in the age group (35,45) tends to take more loans.
- People in the age group (25,35) tends to have high percentage of defaulters.

TOP 10 CORRELATIONS FOR NON-DEFAULTERS ARE-

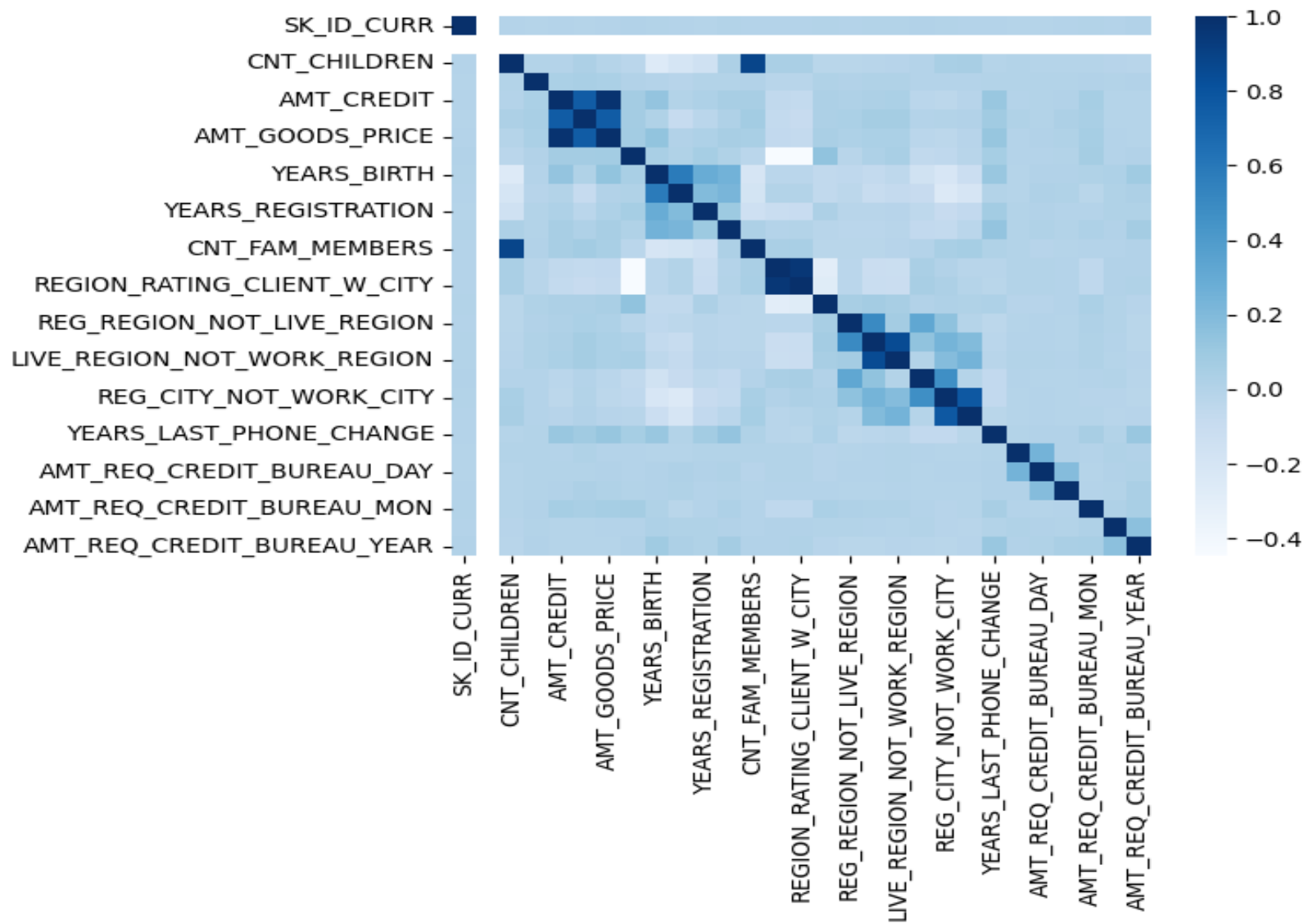
AMT_GOODS_PRICE	AMT_CREDIT	0.986966
REGION_RATING_CLIENT	REGION_RATING_CLIENT_W_CITY	0.950149
CNT_CHILDREN	CNT_FAM_MEMBERS	0.878571
REG_REGION_NOT_WORK_REGION	LIVE_REGION_NOT_WORK_REGION	0.861861
REG_CITY_NOT_WORK_CITY	LIVE_CITY_NOT_WORK_CITY	0.830381
AMT_GOODS_PRICE	AMT_ANNUITY	0.776686
AMT_ANNUITY	AMT_CREDIT	0.771309
YEARS_BIRTH	YEARS_EMPLOYED	0.626028
REGION_RATING_CLIENT	REGION_POPULATION_RELATIVE	0.539005
REGION_POPULATION_RELATIVE	REGION_RATING_CLIENT_W_CITY	0.537301



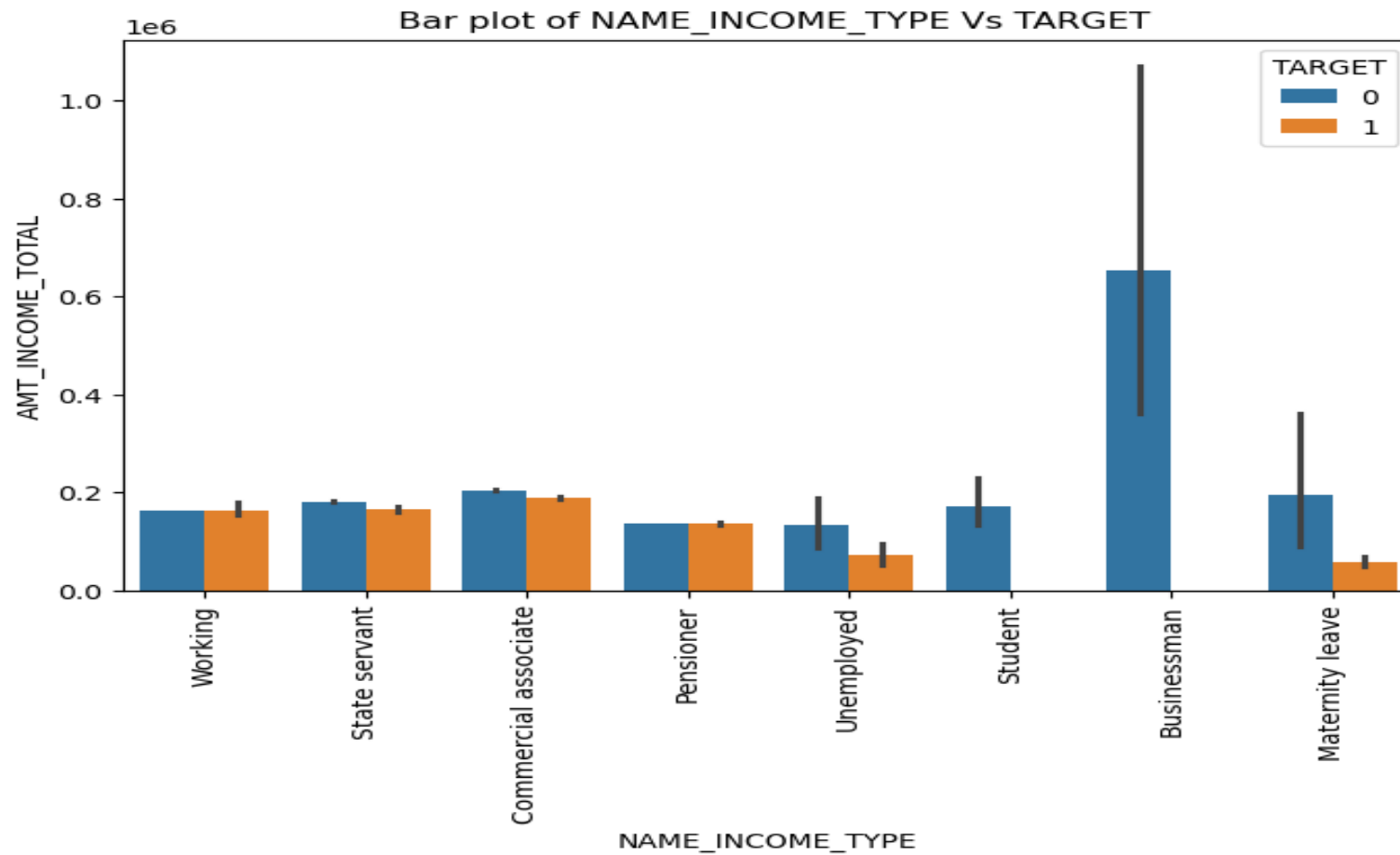
➤ Heat map for Non-defaulters.

Top 10 correlations for Defaulters are:

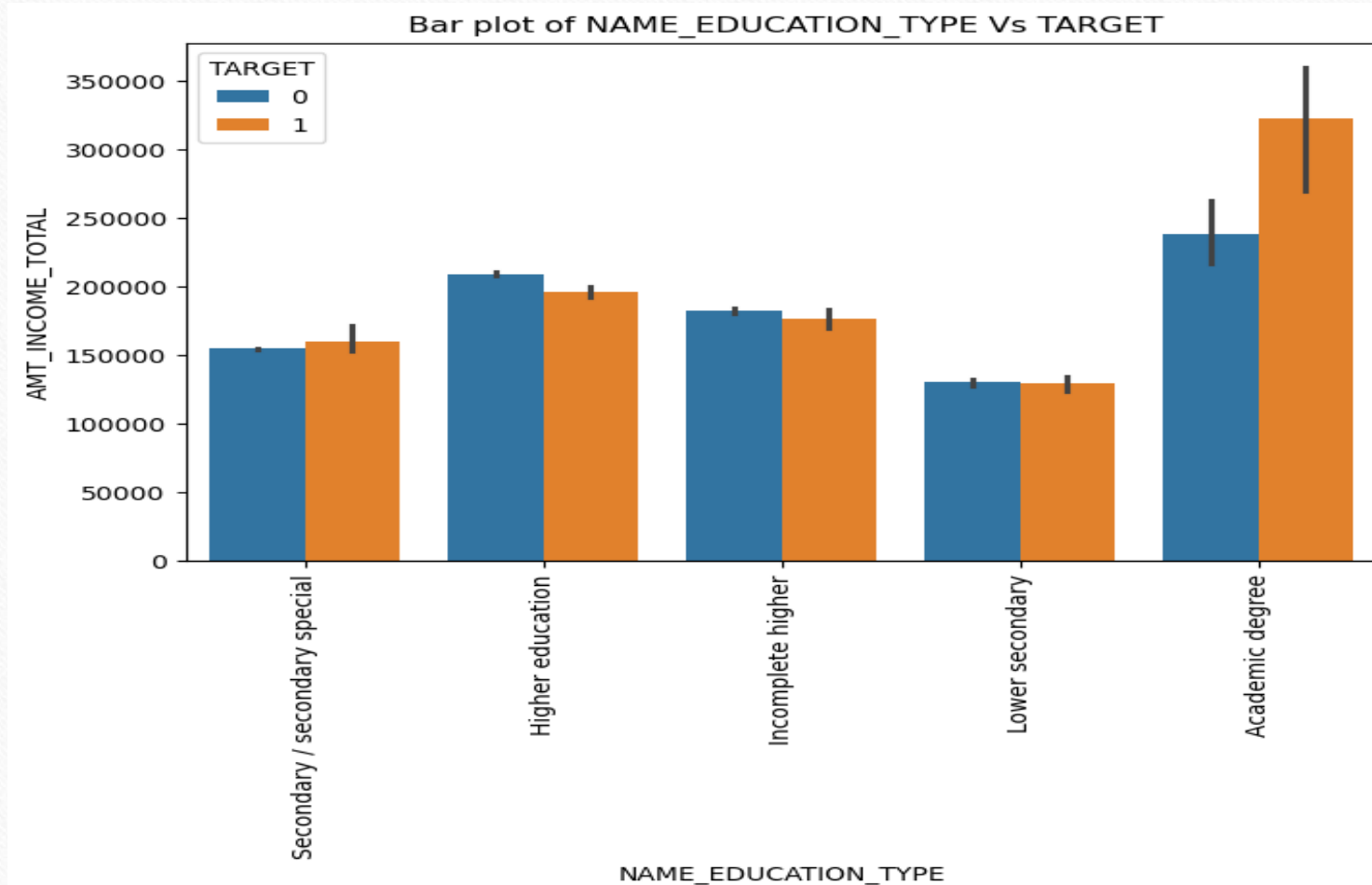
AMT_CREDIT	AMT_GOODS_PRICE	0.982854
REGION_RATING_CLIENT_W_CITY	REGION_RATING_CLIENT	0.956637
CNT_FAM_MEMBERS	CNT_CHILDREN	0.885484
LIVE_REGION_NOT_WORK_REGION	REG_REGION_NOT_WORK_REGION	0.847885
LIVE_CITY_NOT_WORK_CITY	REG_CITY_NOT_WORK_CITY	0.778540
AMT_GOODS_PRICE	AMT_ANNUITY	0.752891
AMT_CREDIT	AMT_ANNUITY	0.752195
YEARS_BIRTH	YEARS_EMPLOYED	0.582443
REG_REGION_NOT_WORK_REGION	REG_REGION_NOT_LIVE_REGION	0.497937
REG_CITY_NOT_LIVE_CITY	REG_CITY_NOT_WORK_CITY	0.472052



➤ Heat map for Defaulters.

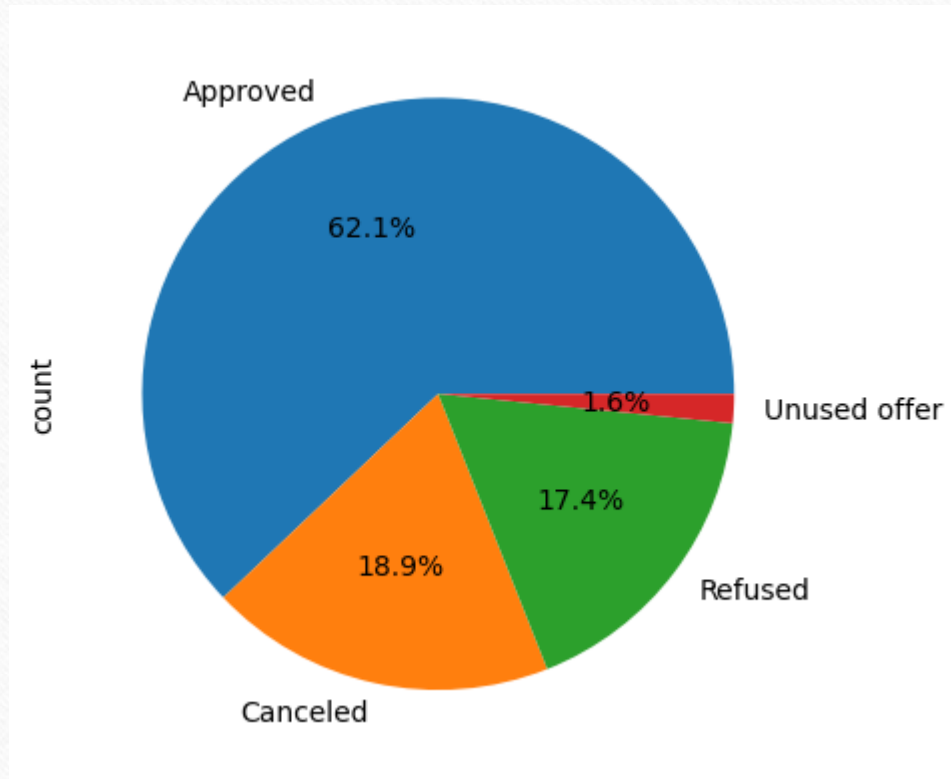


➤ With respect to total income, Business man are non-defaulters.



➤ With respect to total income, Academic degree holder are the highest defaulters.

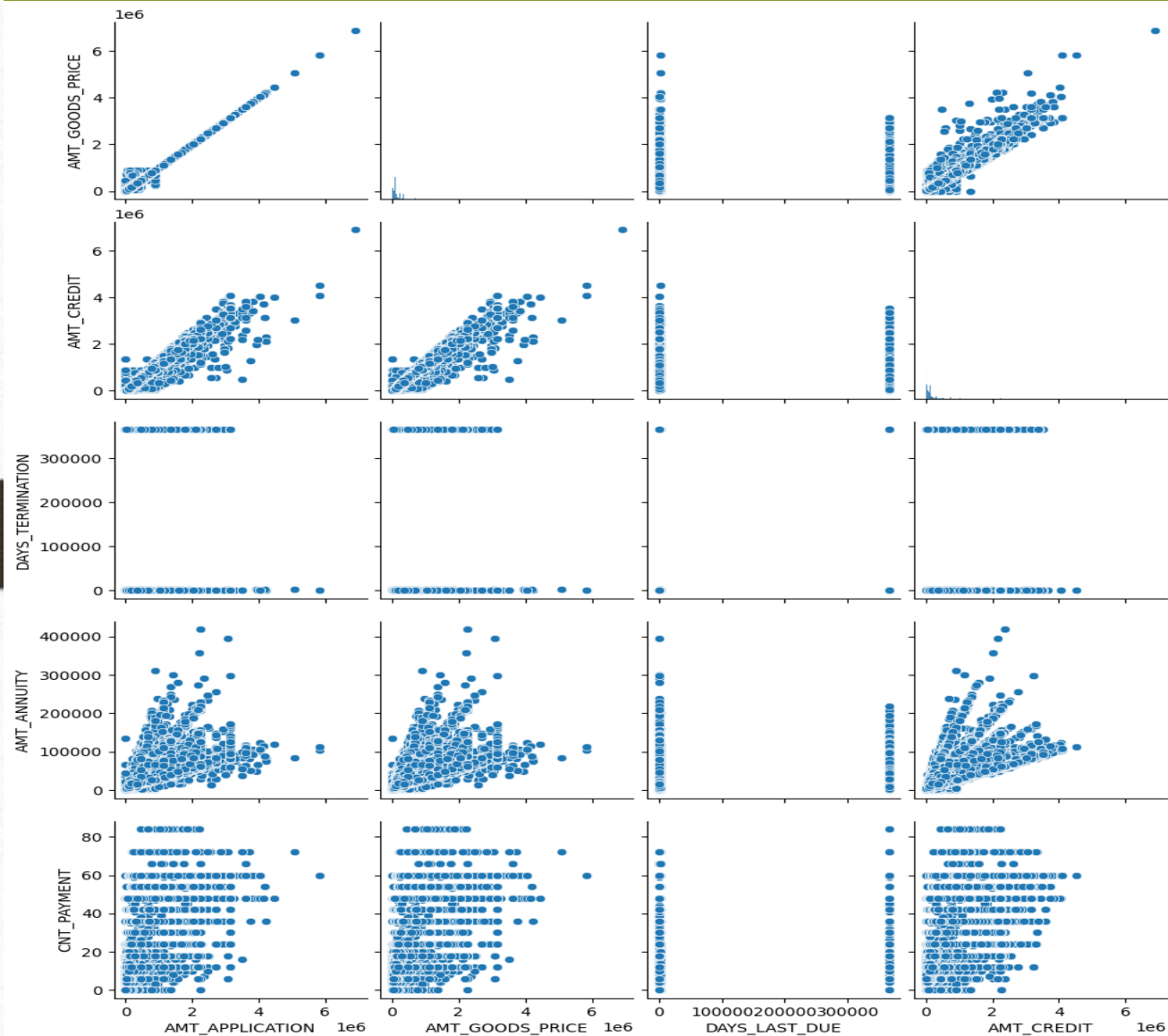
ANALYSIS OF PREVIOUS APPLICATION DATA



- Around 62.1% loan applications were approved.

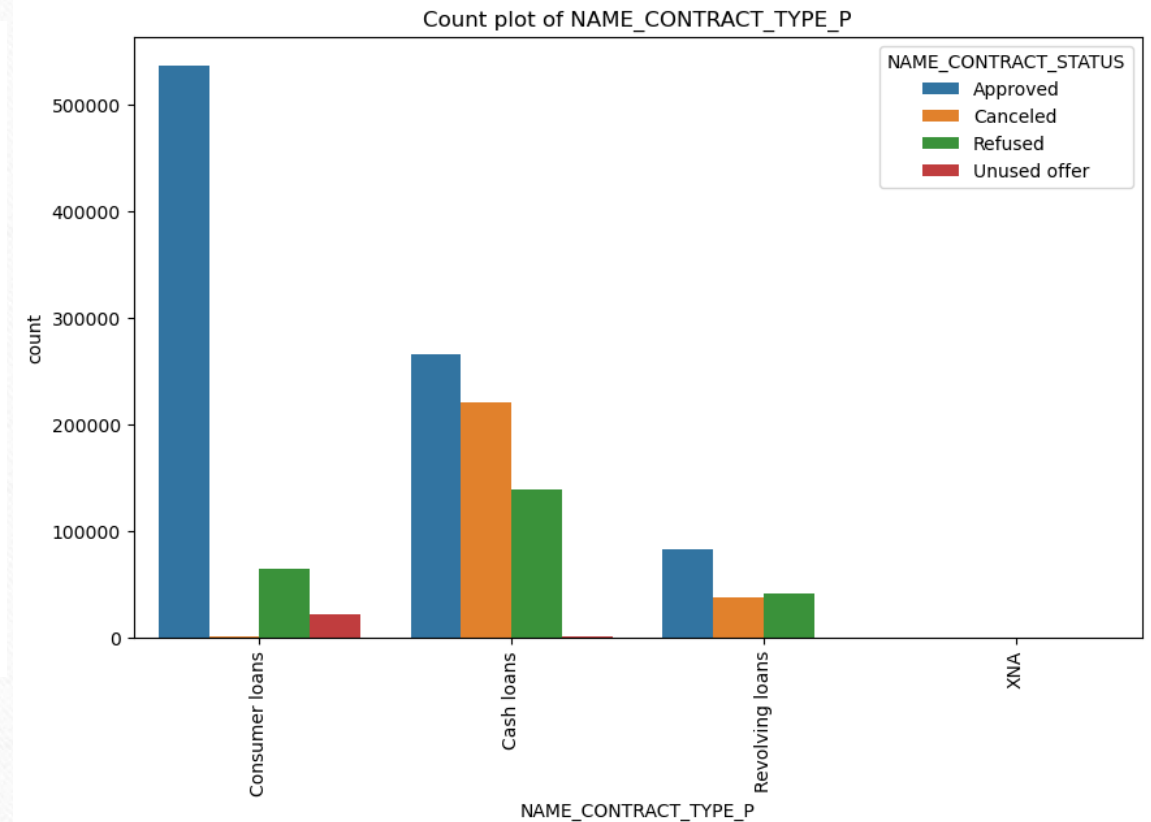
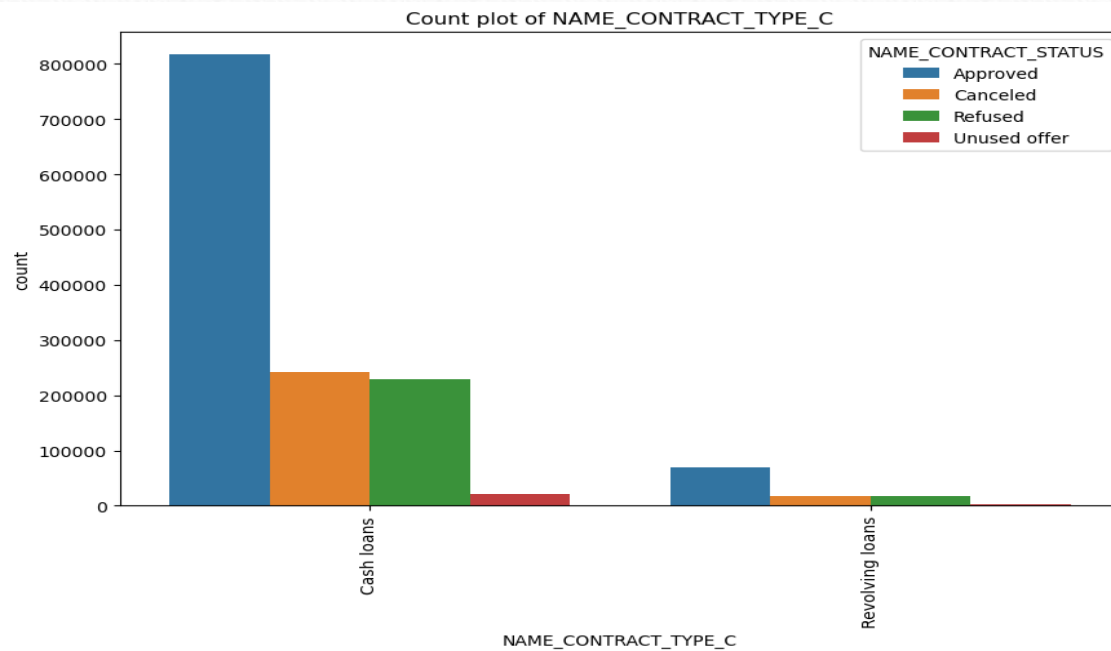
TOP 10 CORRELATION OF PREVIOUS APPLICATION DATA

AMT_APPLICATION	AMT_GOODS_PRICE	0.999884
AMT_GOODS_PRICE	AMT_CREDIT	0.993087
AMT_APPLICATION	AMT_CREDIT	0.975824
DAYS_LAST_DUE	DAYS_TERMINATION	0.927777
AMT_GOODS_PRICE	AMT_ANNUITY	0.820895
AMT_CREDIT	AMT_ANNUITY	0.816429
AMT_APPLICATION	AMT_ANNUITY	0.808872
DAYS_LAST_DUE_1ST_VERSION	DAYS_FIRST_DRAWING	0.803503
AMT_APPLICATION	CNT_PAYMENT	0.680630
AMT_CREDIT	CNT_PAYMENT	0.674278

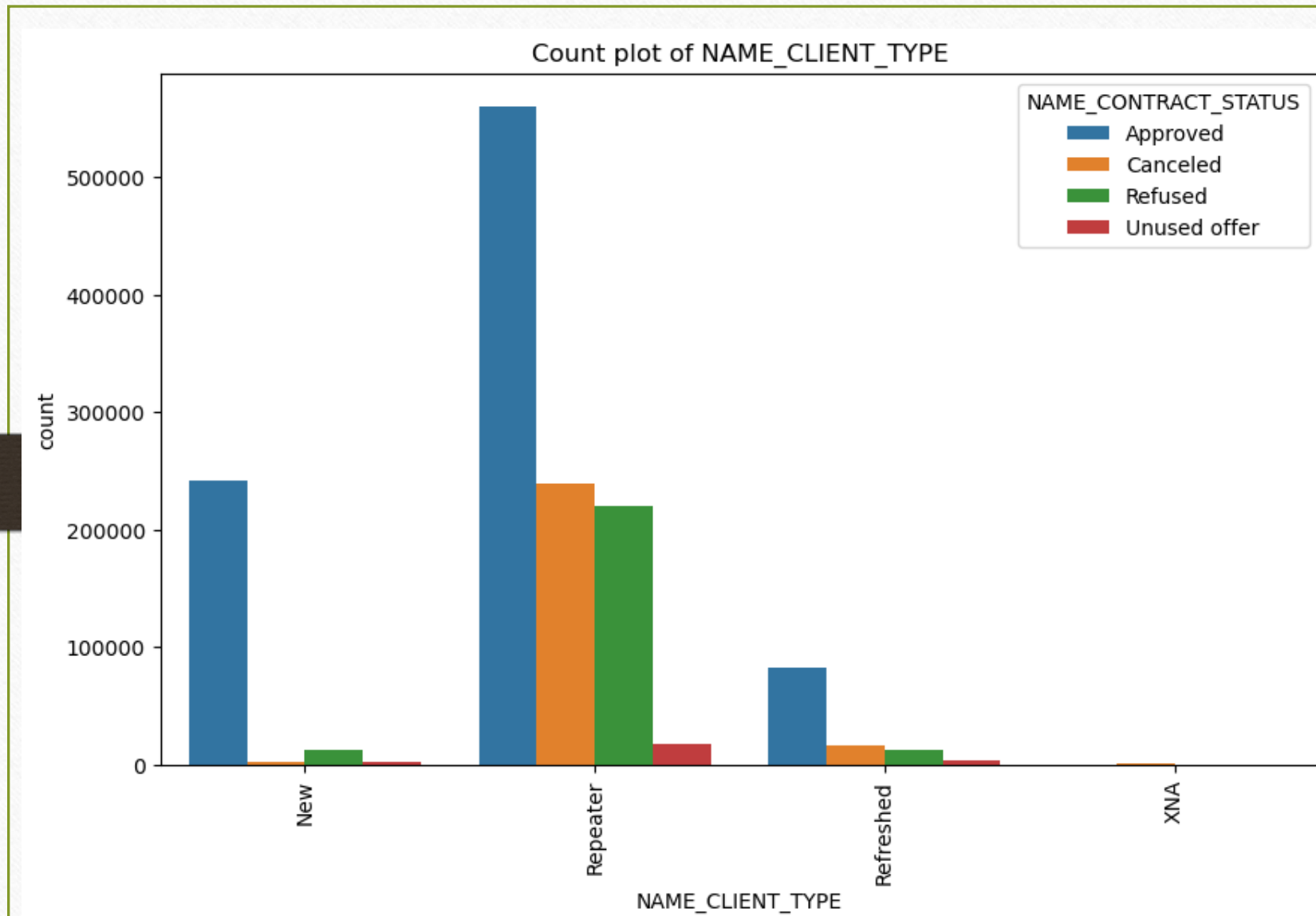


- This pair plot shows the top correlations of previous application data.

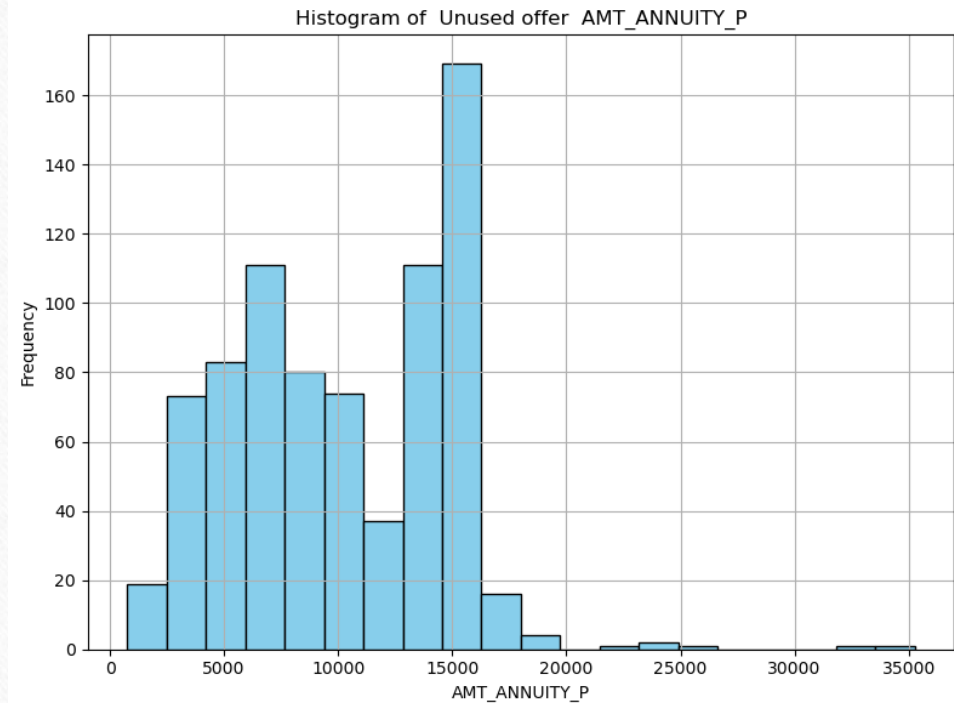
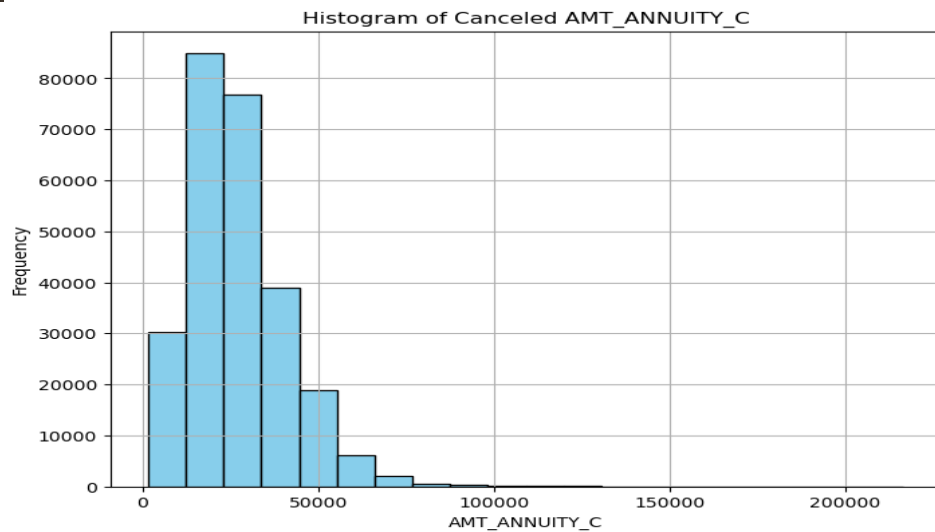
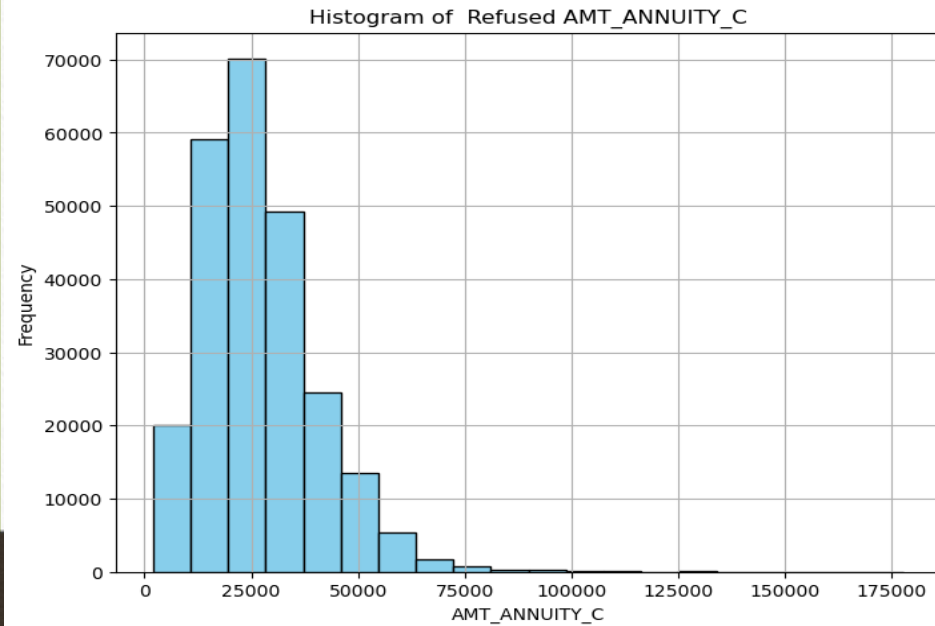
ANALYSIS OF MERGED DATA



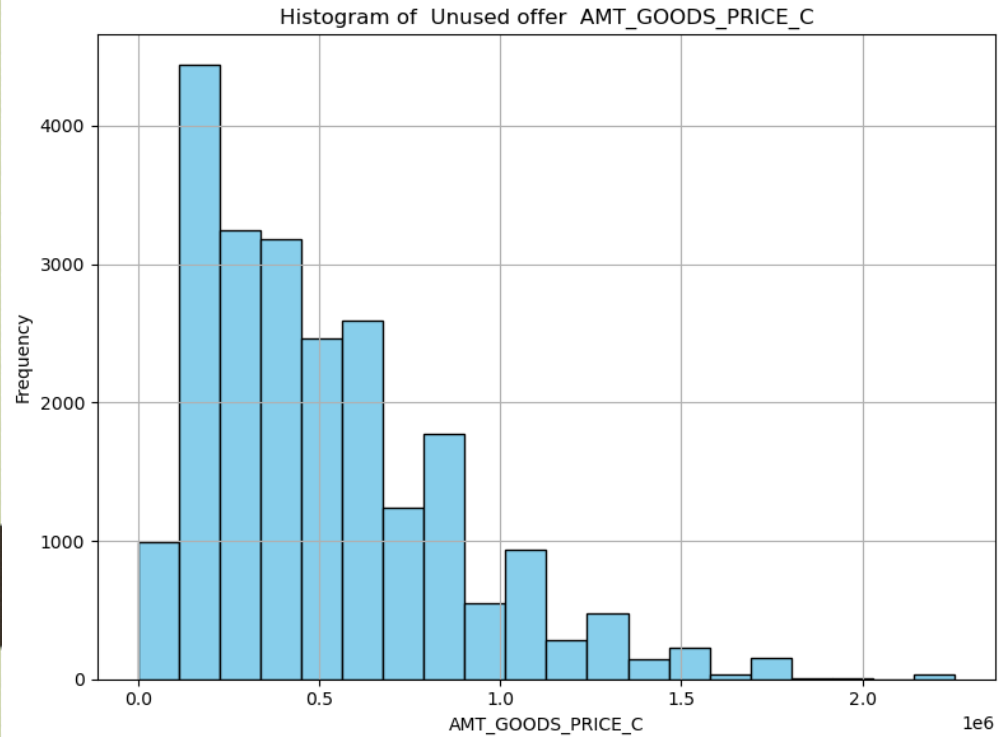
- Number of approved cash loans are highest in current application whereas, in previous application consumer loans were highest.



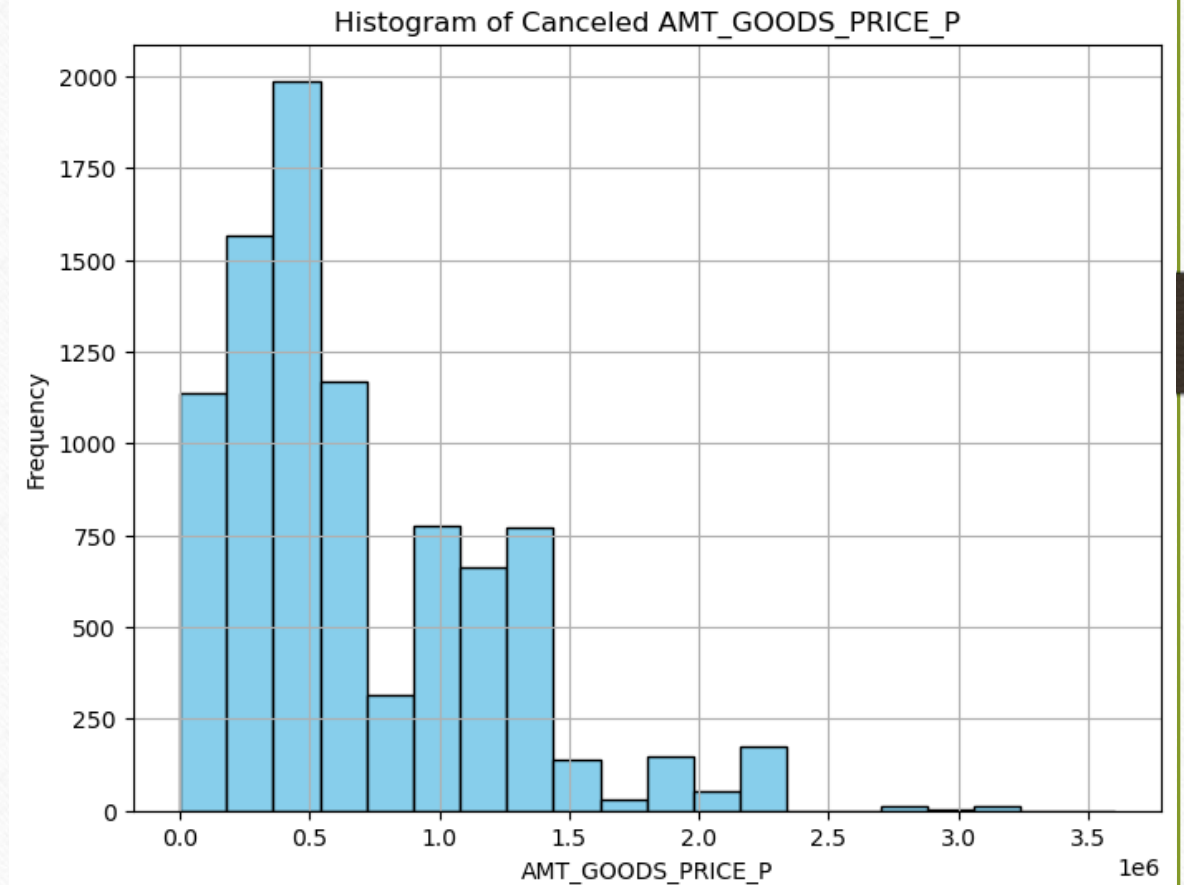
➤ Repeaters has highest number of approved loans.

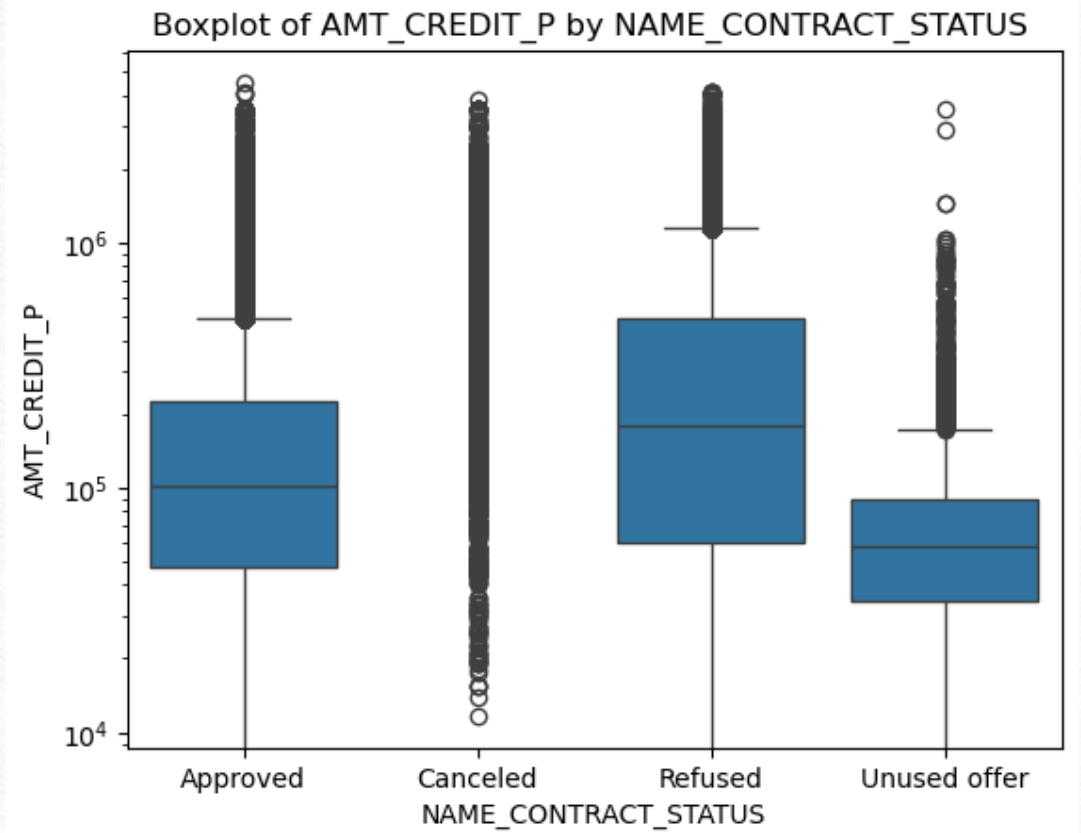
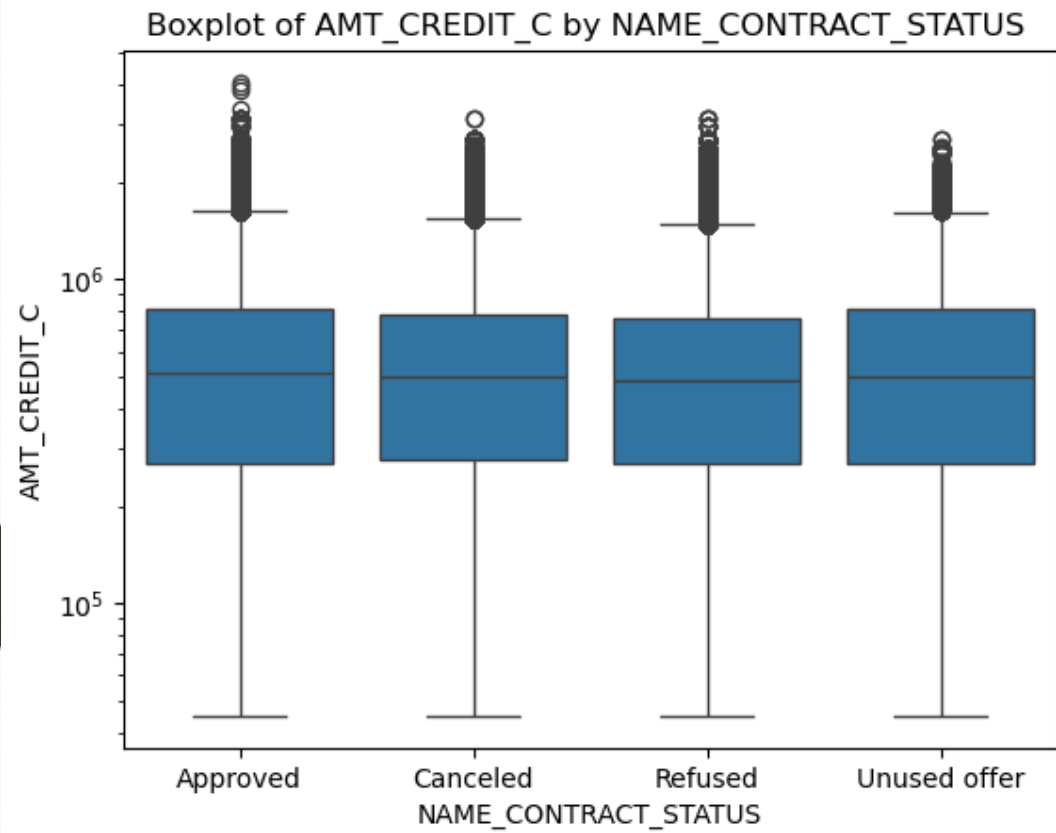


➤ For AMT_ANNUIITY previously bank had high unused offers and currently canceled and refused offers are similar.

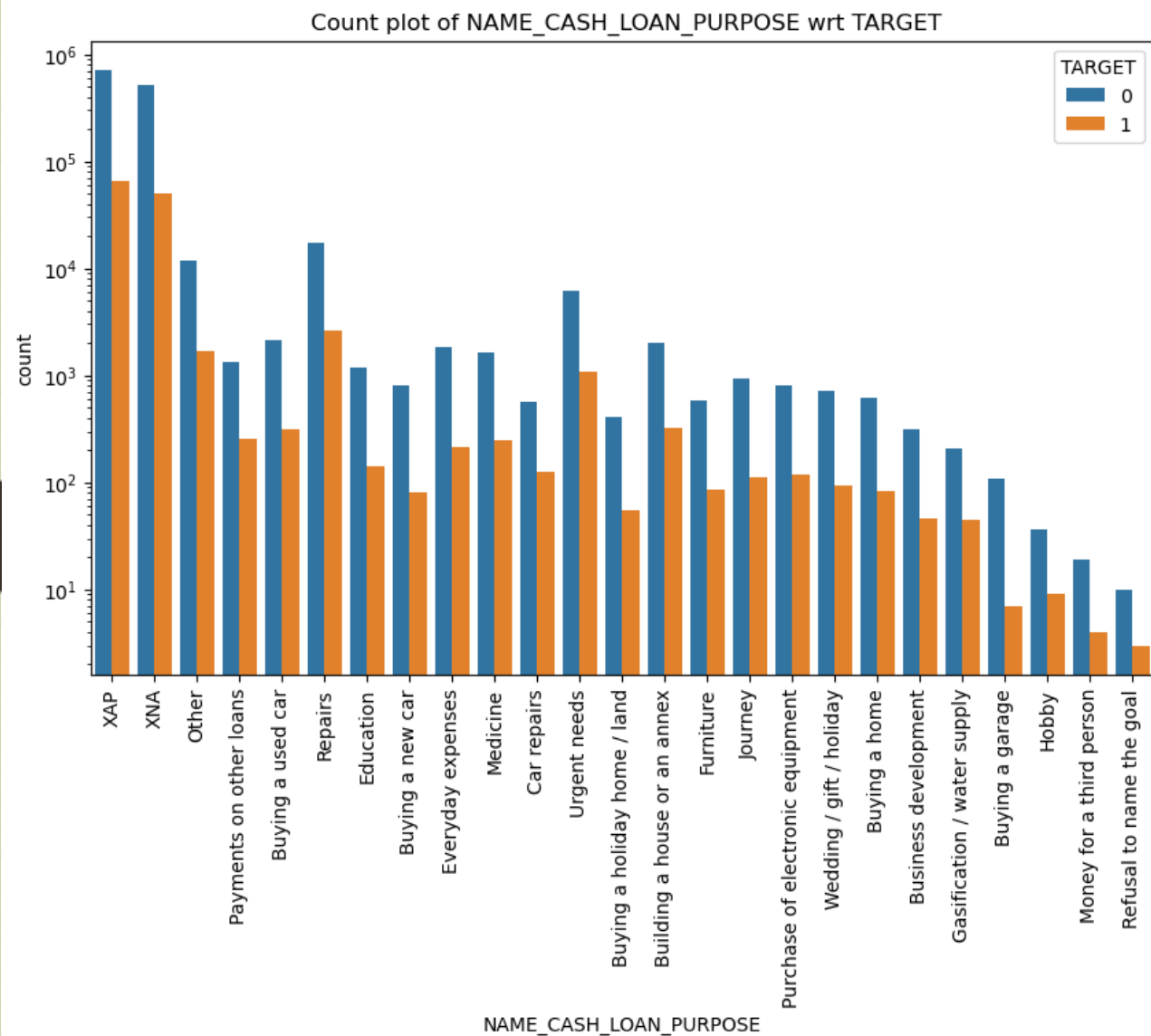


➤ For AMT_GOODS_PRICE, currently bank had high unused offers and previously canceled is high.

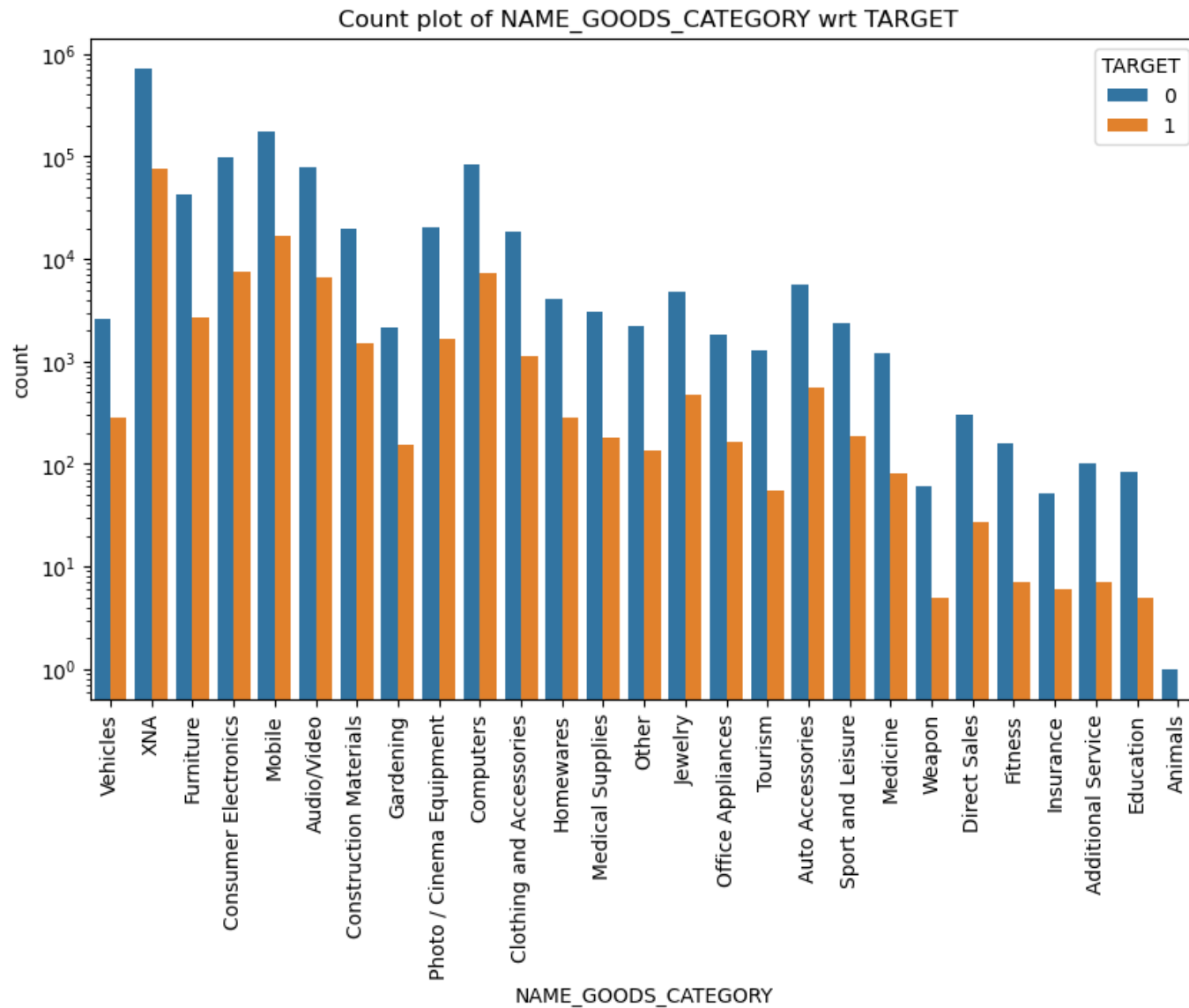




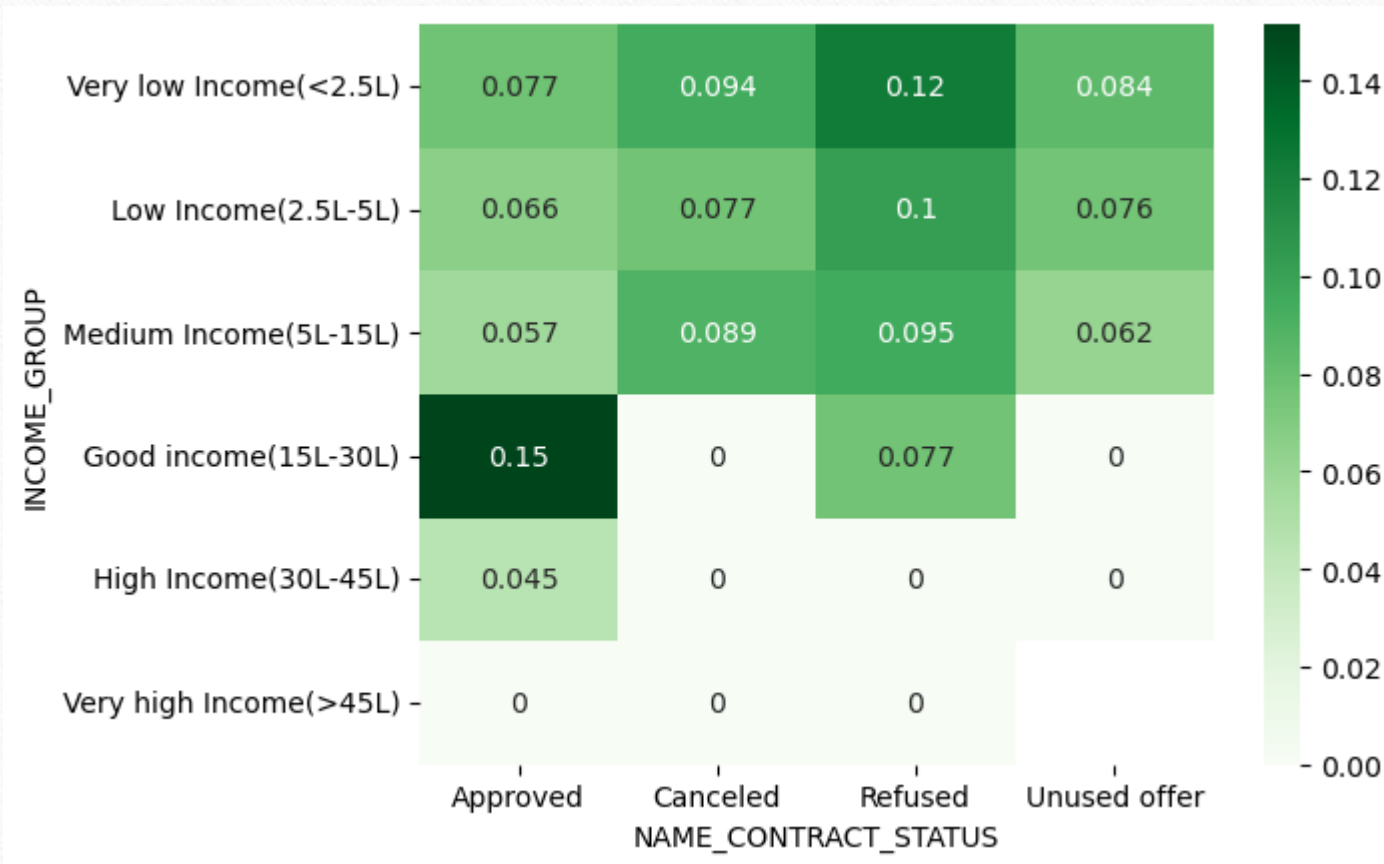
- AMT_CREDIT current is similar for all cases whereas AMT_CREDIT previous has highest refused cases.



- In non-default cases, most of the people uses the loan amount for repairing.



➤ For non-defaulters, most of the people applied for mobiles in goods category.



➤ Good income(15L-30L) has highest approvals.



- Highest number of refused cases are from age group (15,25) and (25,35)

CONCLUSION

The variables to be considered before approving loan applications to minimize risk of loss are:

- Gender of the applicants. (CODE_GENDER)
- Income of the applicants. (AMT_INCOME_TOTAL)
- Age of the applicants. (YEARS_BIRTH)
- Education of the applicants. (NAME_EDUCATION_TYPE)
- Loan amount. (AMT_ANNUITY)
- Credit amount of the loan. (AMT_CREDIT)
- Rating of the region in which the applicant reside. (REGION_RATING_CLIENT)
- Number of family members of the applicants. (CNT_FAM_MEMBERS)
- Housing situation of the applicant. (NAME_HOUSING_TYPE)
- Applicant income type. (NAME_INCOME_TYPE)

THANK YOU !!